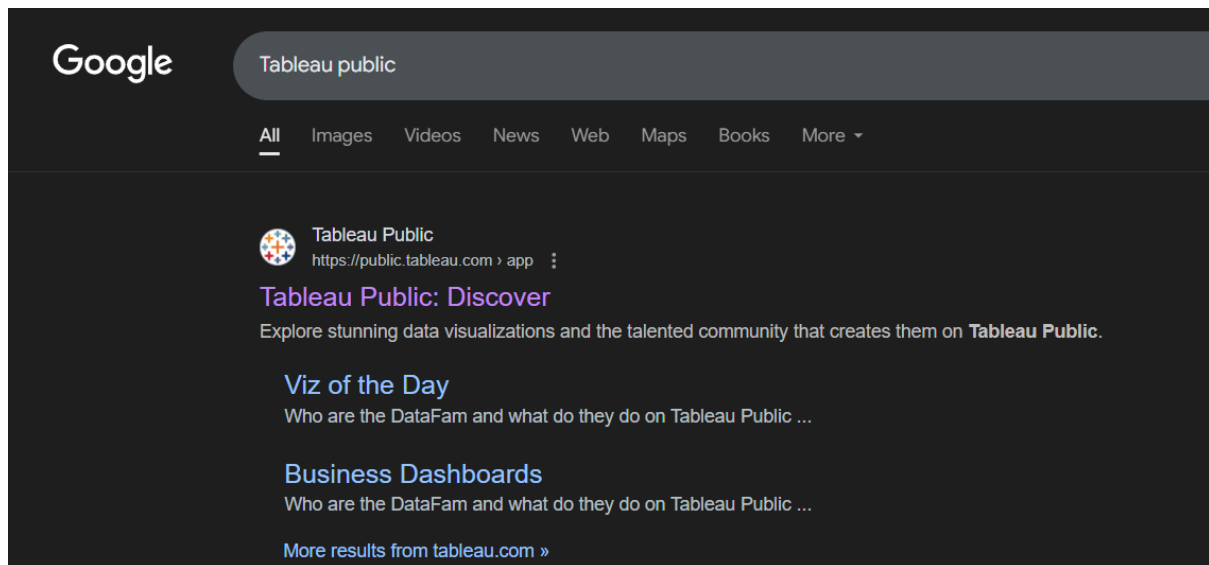
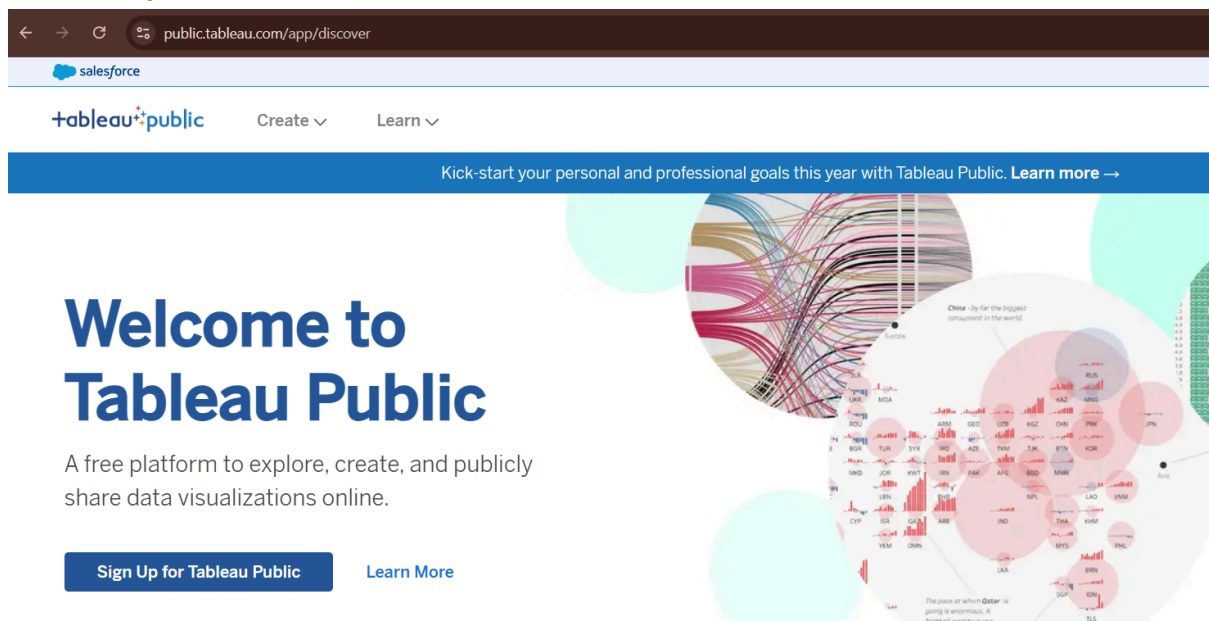


Basic Installation Steps
Search Tableau public on google.



Click on Sign Up for Tableau Public.



Then create your Tableau account.

Create your Tableau Account

Passwords are case sensitive, 8-100 characters long. Requires at least one number (0-9), one uppercase letter, one lowercase letter, and one of the following characters: ! " \ # \$ % & ' () * + , - . / : ; < = > ? @ [] ^ _ ` { | } ~.

CREATE MY ACCOUNT

ALREADY HAVE AN ACCOUNT? [SIGN IN](#)

After creating the account Sign In.

Sign In

☐ Remember me

[SIGN IN](#)

[FORGOT PASSWORD](#) | [CREATE AN ACCOUNT](#)

Then Download Tableau Desktop Public Edition.

← → ↺ 🔍 public.tableau.com/app/discover

 salesforce

tableau + public

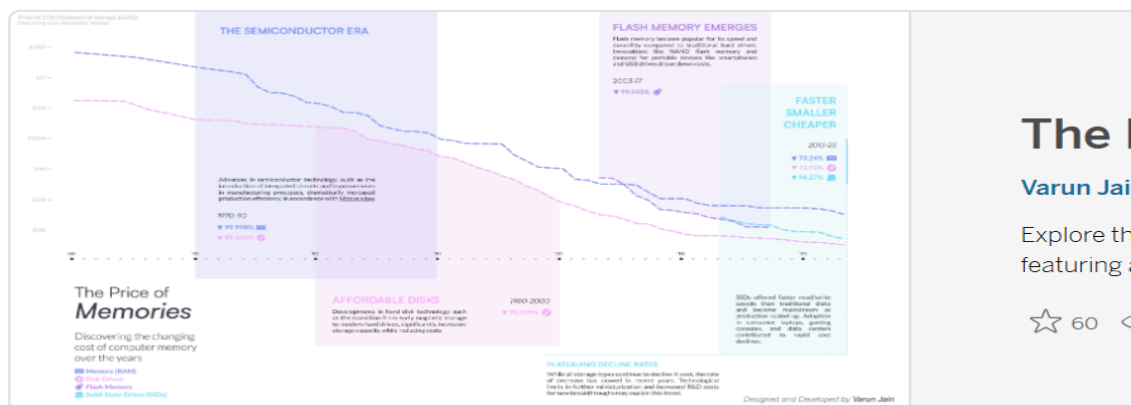
Create ▾ Learn ▾

Web Authoring

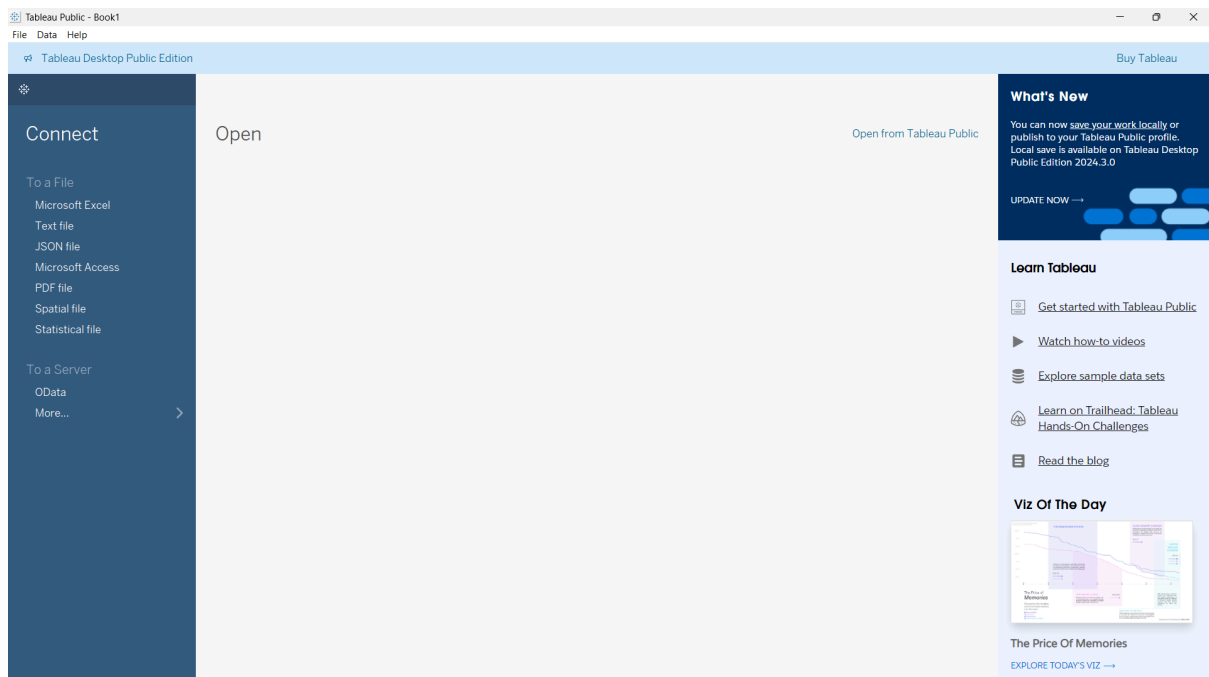
Download Tableau Desktop Public Edition

Viz of the Day [+ Subscribe](#)

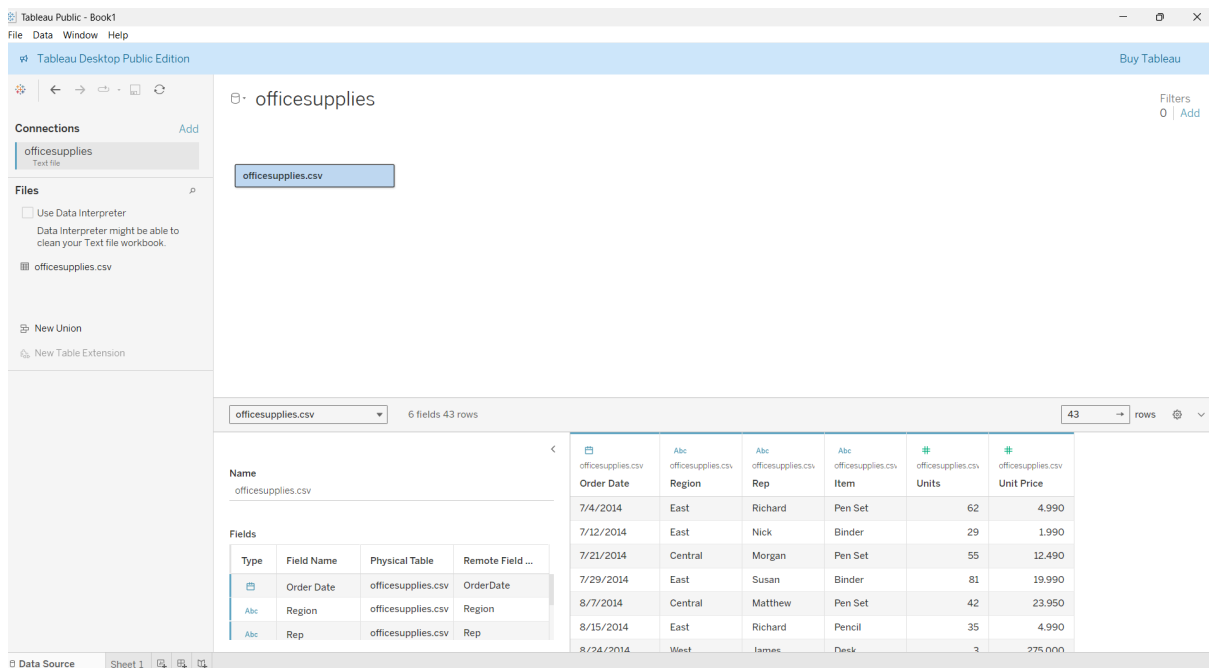
See new and inspiring data visualizations featured every weekday



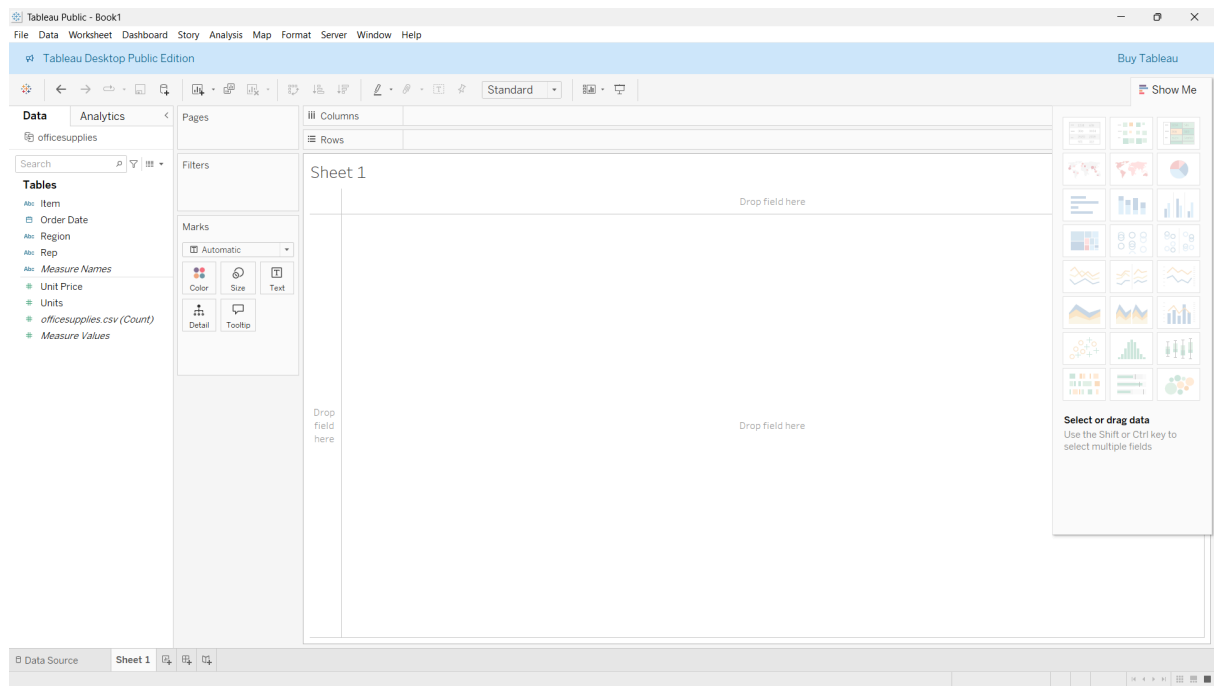
After downloading is completed, install and open the application.



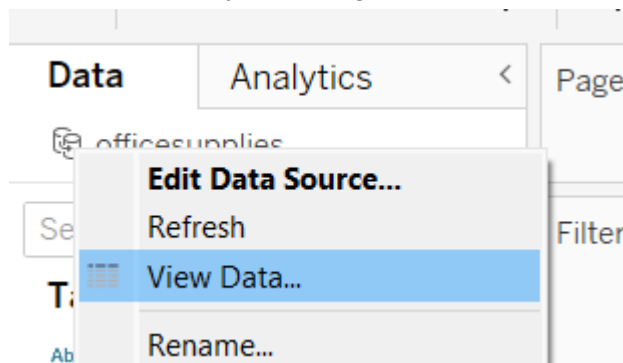
1) Connecting Tableau to the data file.
Click on Text File and Import the officesupplies.csv file.



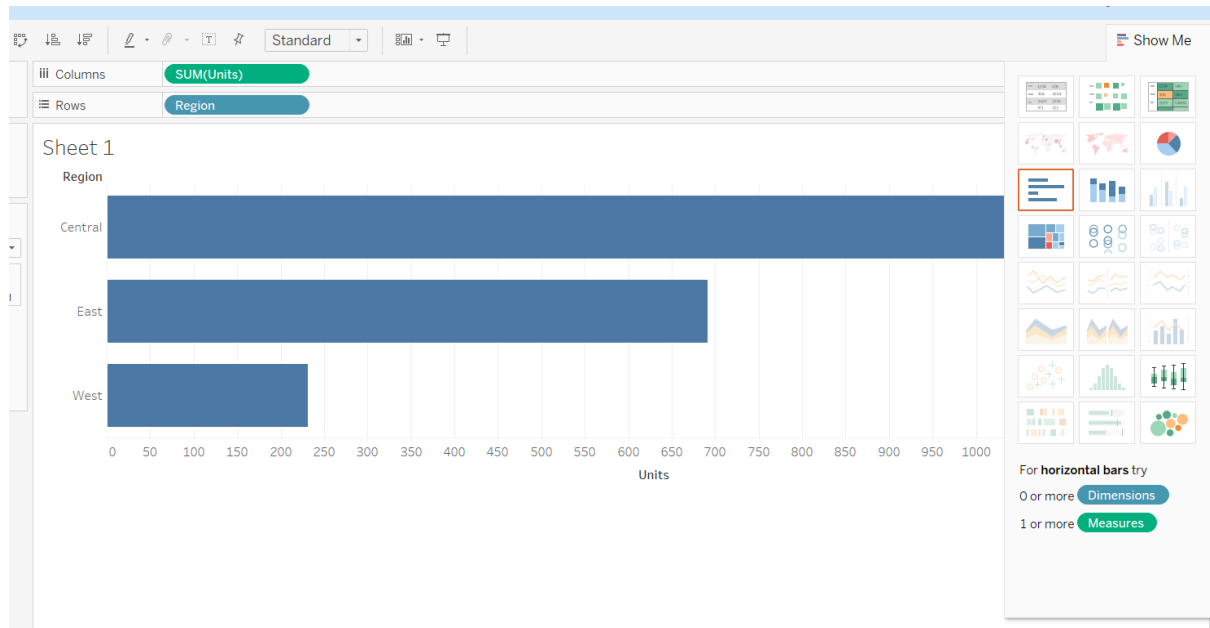
We can go to the dashboard by clicking on Sheet 1.



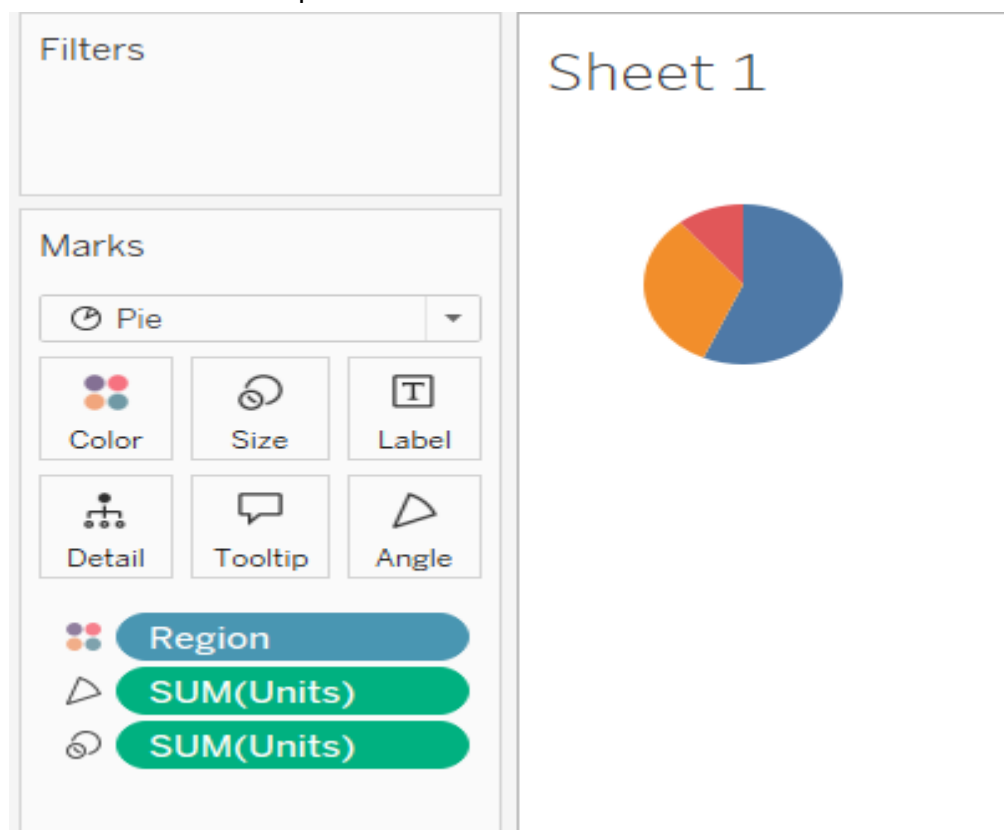
To view the data you can right click on officesupplies and then on view data.



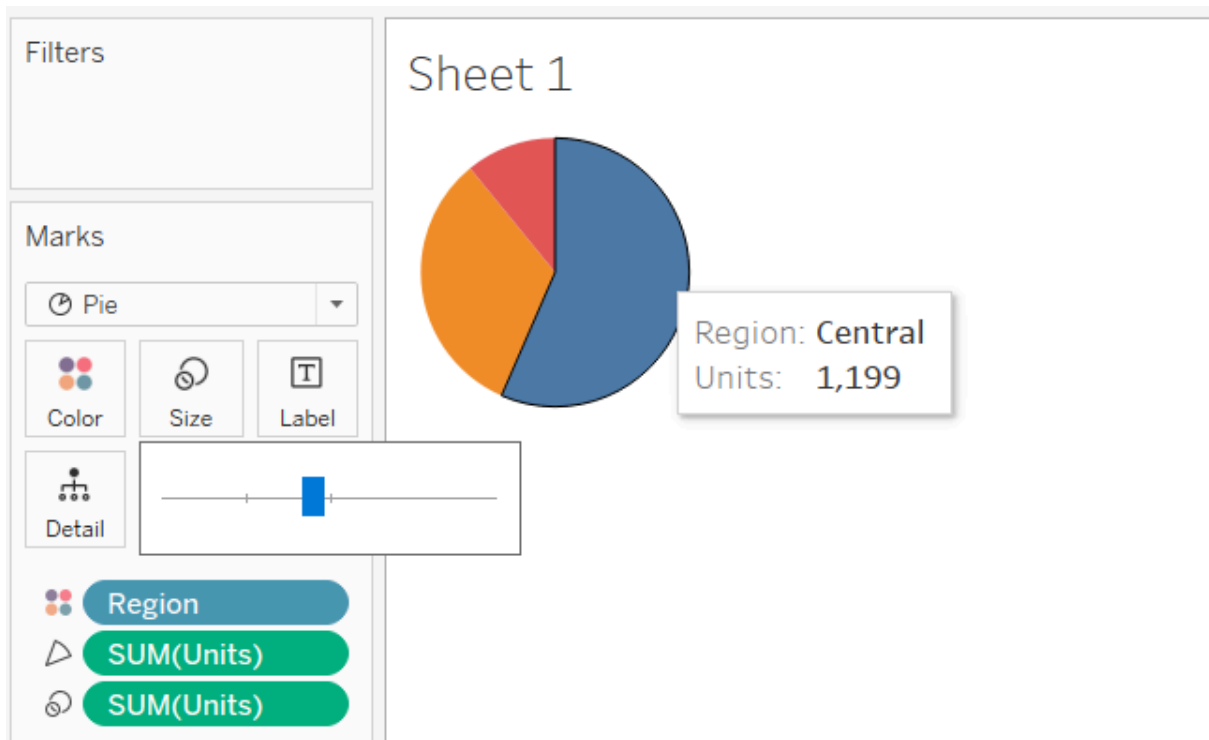
Right away we have a bar chart and now we can extend or reduce the size of it.



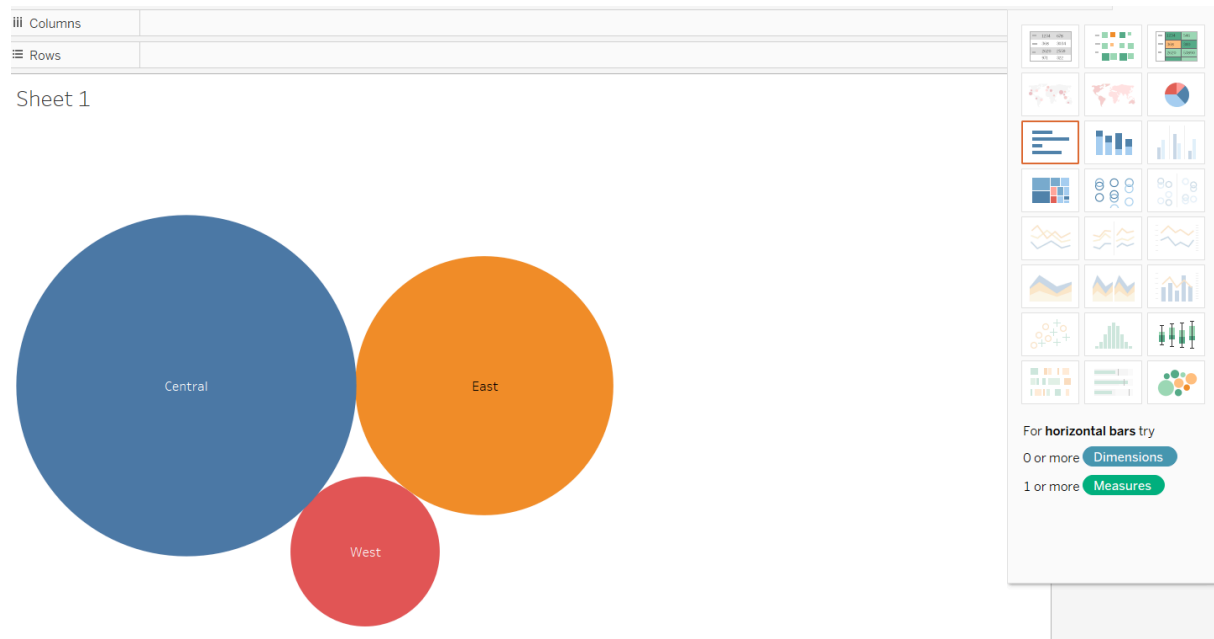
You can also change the type of the chart. For instance, let's display it as a pie chart. Click on Show Me and then pie-chart.



You can change the size of the chart and hover on the part of which information you want.



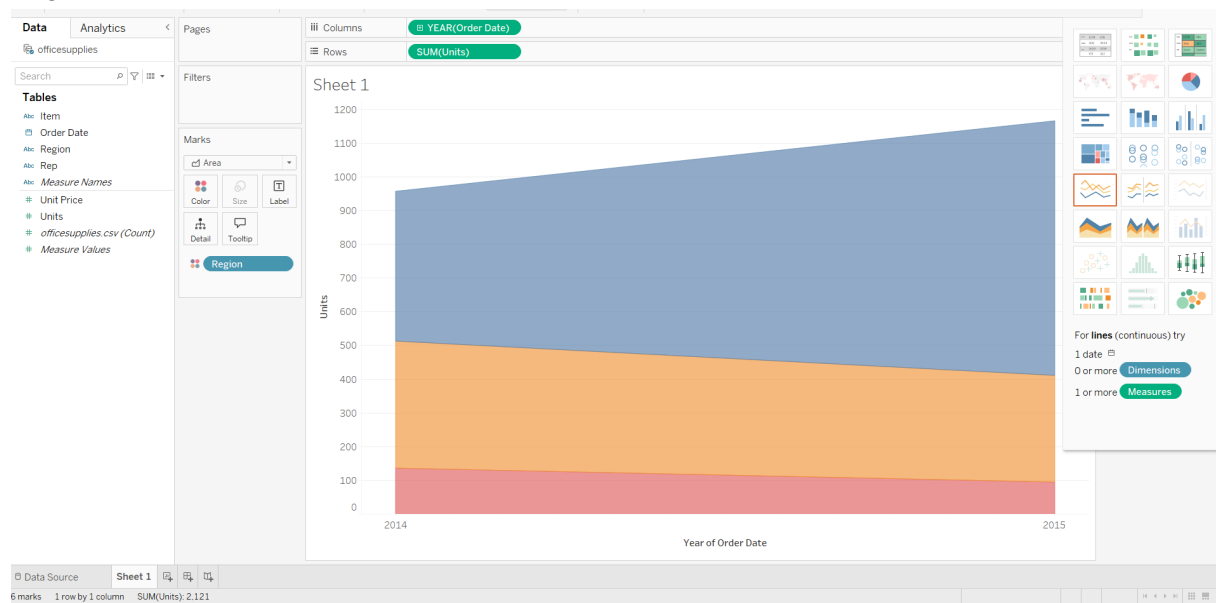
You can also change it to the bubble chart.



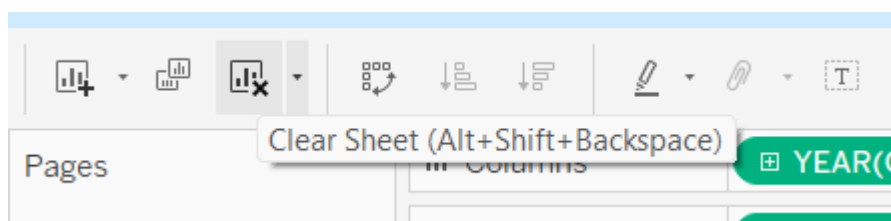
You can also change it to treemap.



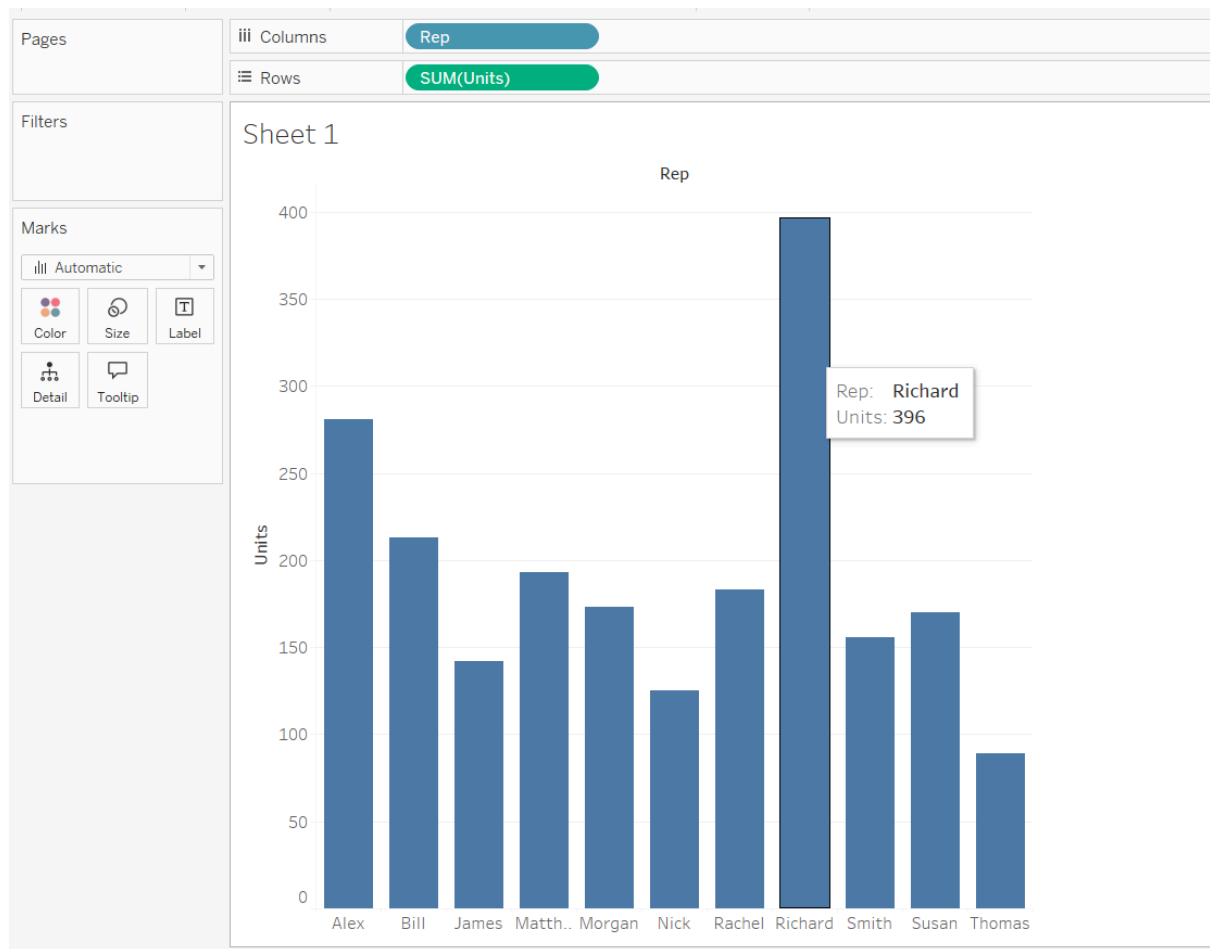
To create an area chart, you need at least one date.
So, drag and drop Order Date on the Columns.
Region on rows and Year(Order Date) on columns.



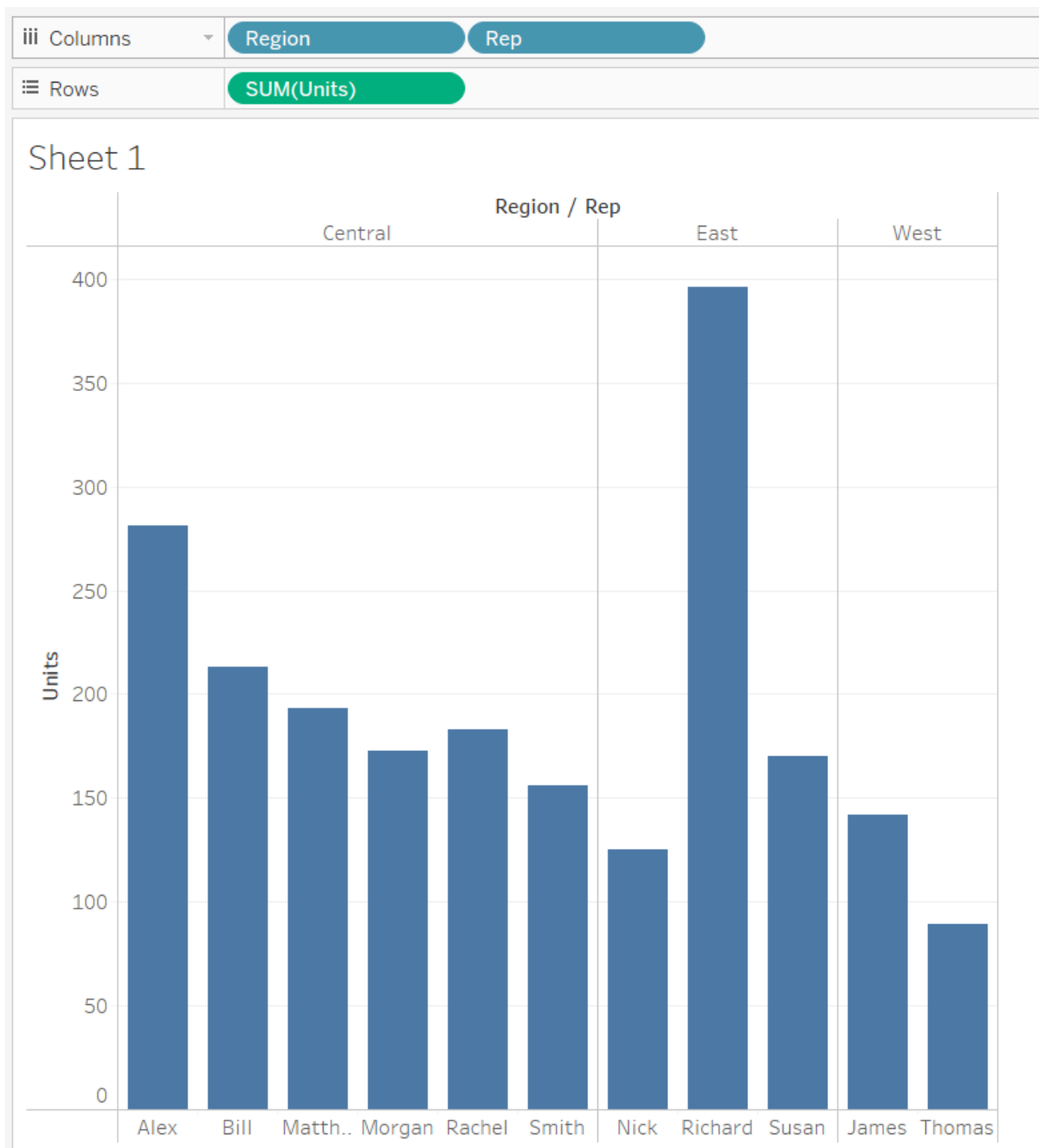
3) Creating Calculated Fields
Clear the sheet.



Drag and Drop Rep to the Columns and Units to the Rows.
We can see that Richard has sold the most Units.

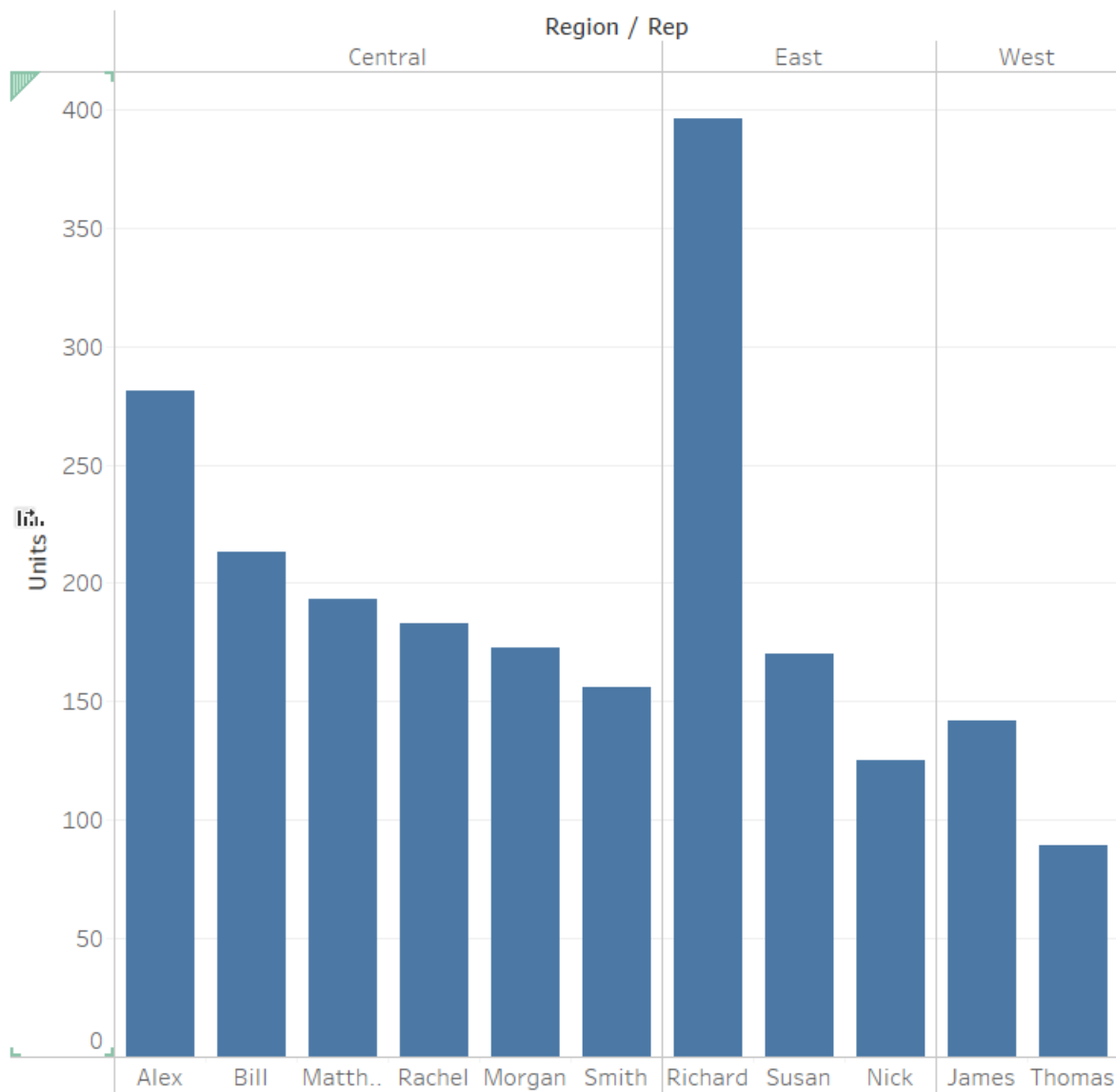


Best Representative in each region?
Drag Region in the Columns but put it before Rep.



We can order these bars by their size.
Click on the button above Units to order the bars.

Sheet 1



Units is just only number of items sold but the total price is Unit Price * Units.
Right click on measures and click on Create Calculated Field.

Data

Analytics

Pages

officesupplies

Search

Filters

Tables

Item

Order Date

Region

Rep

Measure Names

Unit Price

Units

officesupplies.csv (Count)

Measure Values

Marks

Automatic

Color

Size

Label

Detail

Tooltip

Create Calculated Field...

Create Parameter...

Create Folder (use group by folder)

Group by Folder

•

Group by Data Source Table

•

Sort by Name

Sort by Data Source Order

Hide All Unused Fields

Show Hidden Fields

Expand All

Collapse All

Rename it as TotalSales and Multiply Unit Price and Units.
Then press OK and New measure TotalSales is created.

TotalSales

×

[Unit Price] * [Units]|

►

The calculation is valid.

Apply

OK

Tables

Item

Order Date

Region

Rep

Measure Names

TotalSales

Unit Price

Units

officesupplies.csv (Count)

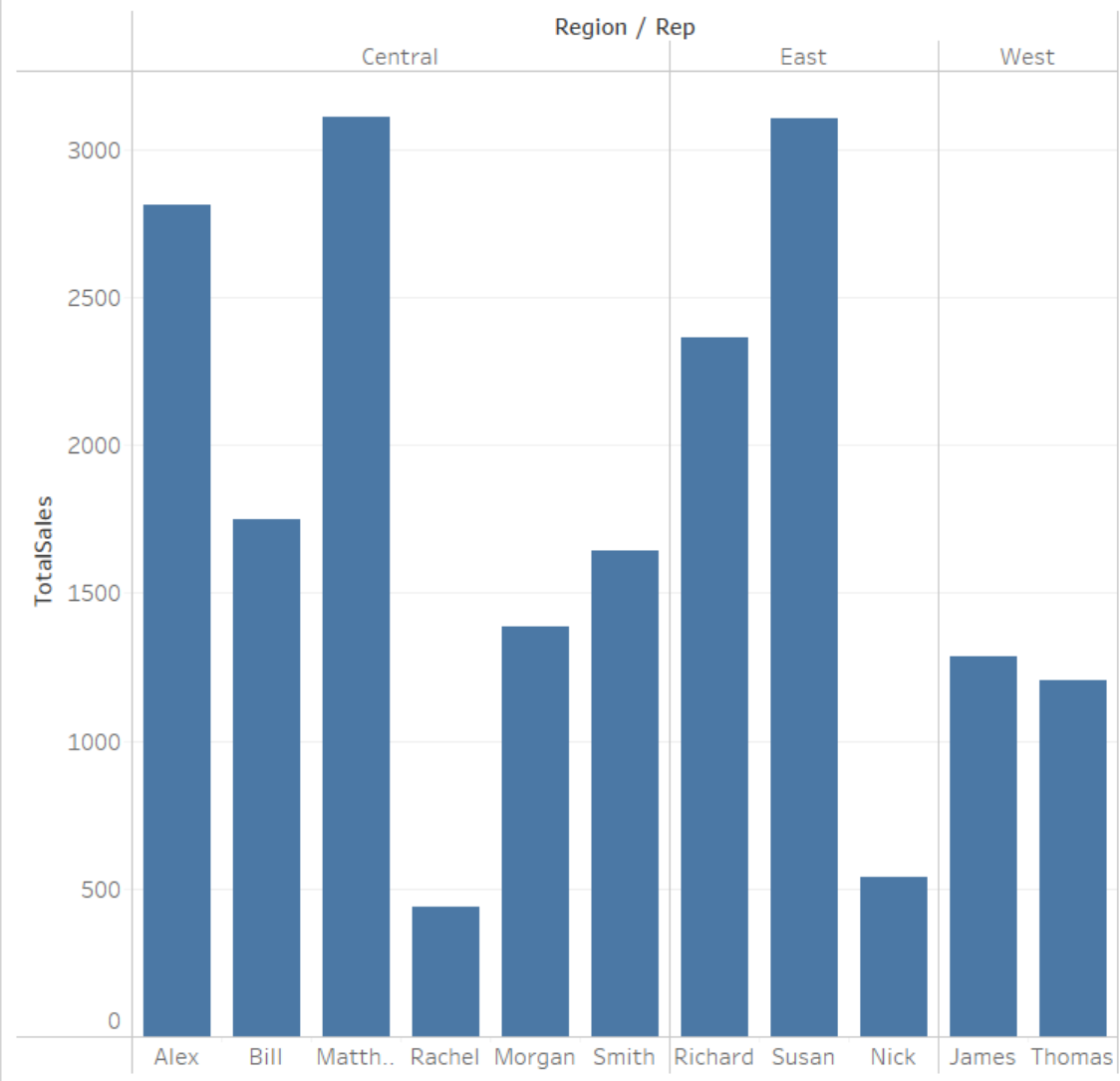
Measure Values

Now, drag and drop TotalSales in Rows to replace SUM(Units).
Drag it exactly on top.

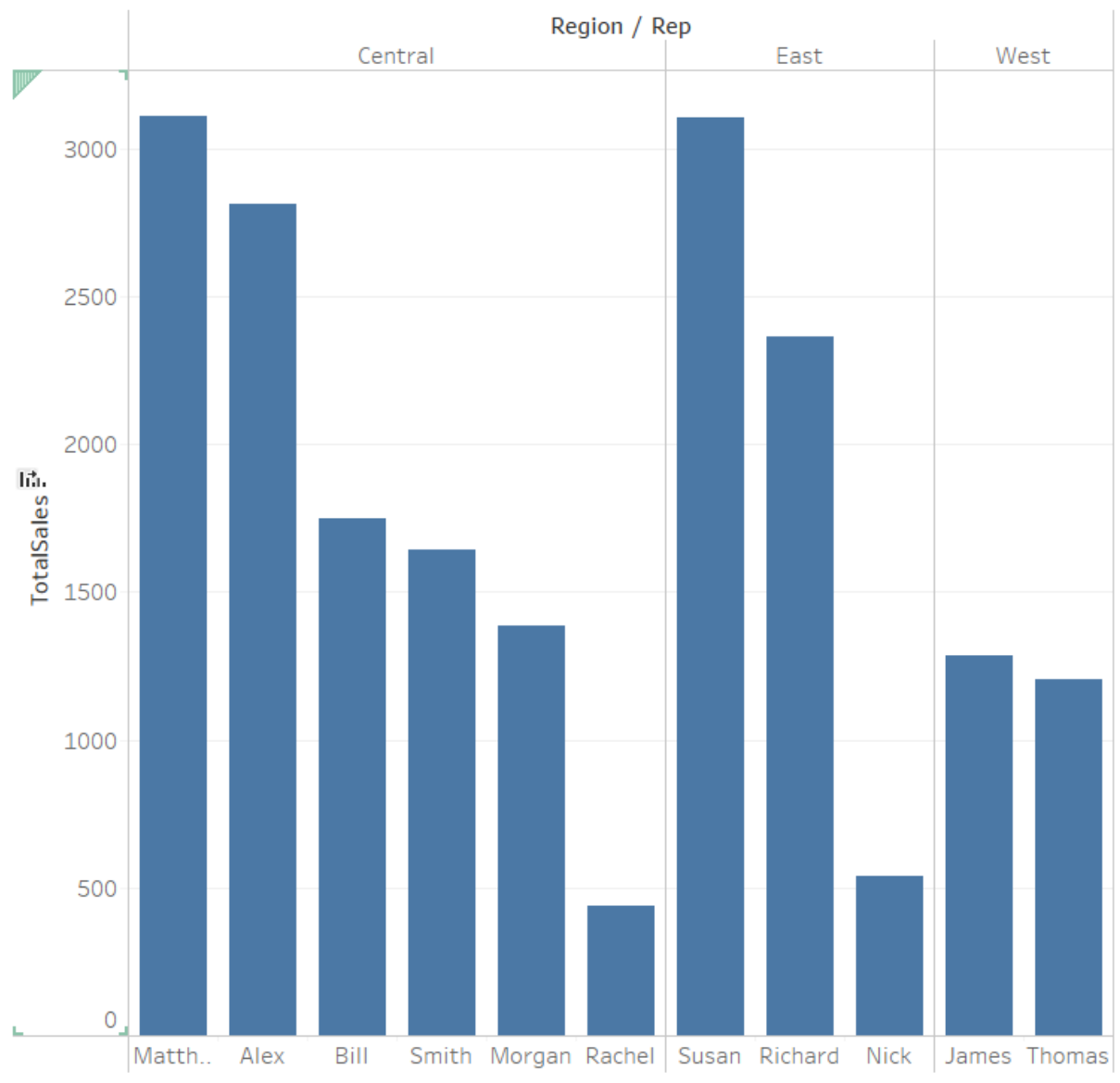
Columns	Region	Rep	≡
Rows	SUM(TotalSales)		

Columns	Region	Rep
Rows	SUM(TotalSales)	

Sheet 1

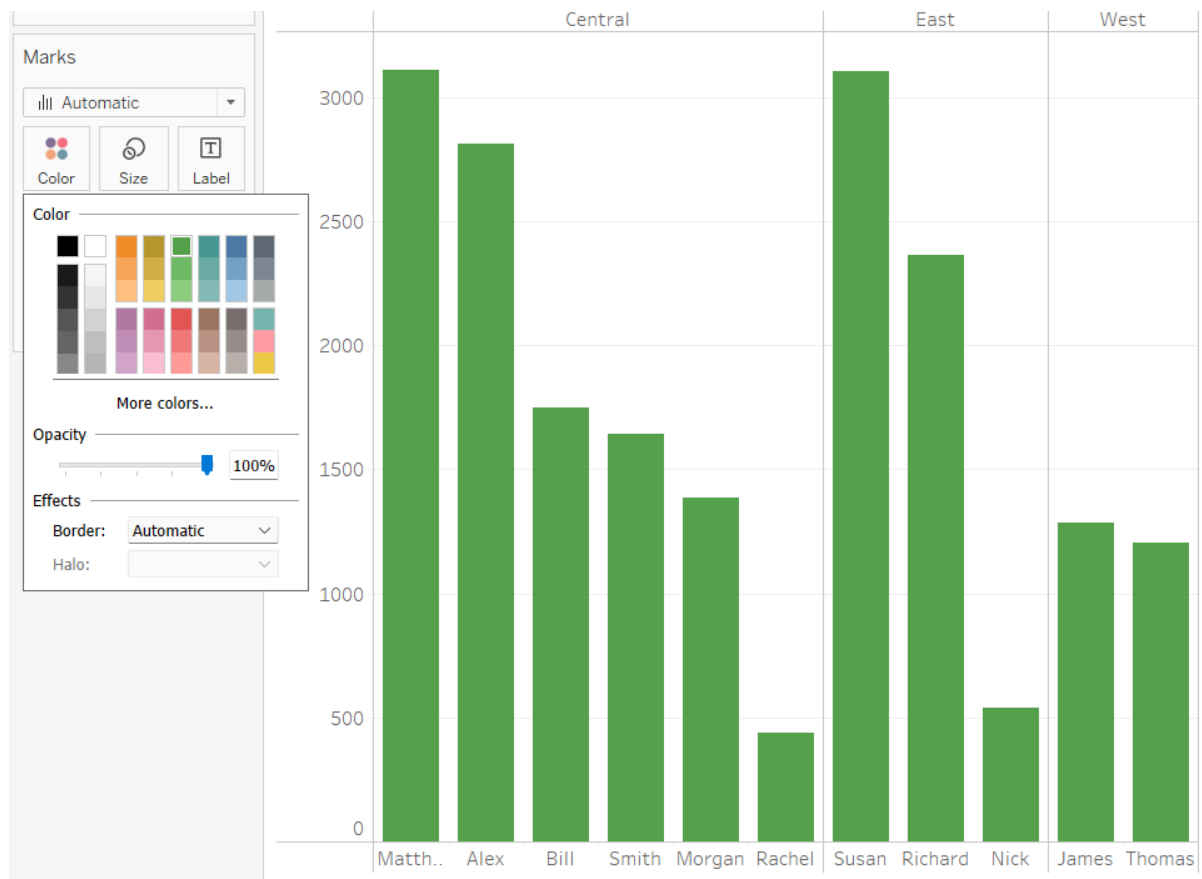


You can order by value by clicking Total Sales and find out who deserved the bonus.

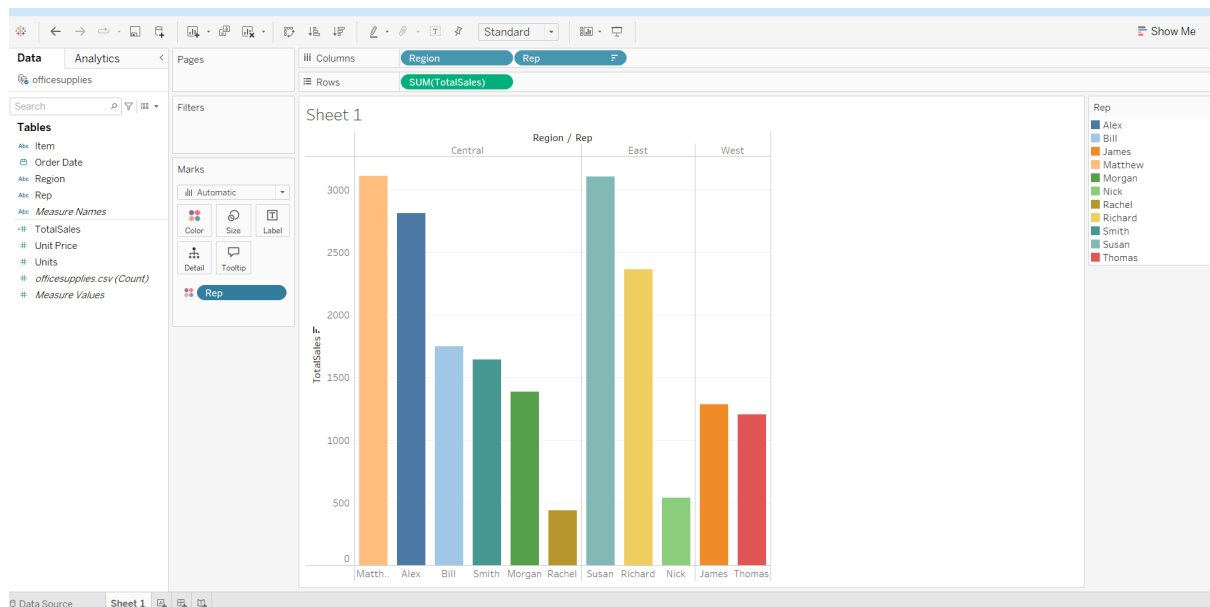


4) Adding Colors

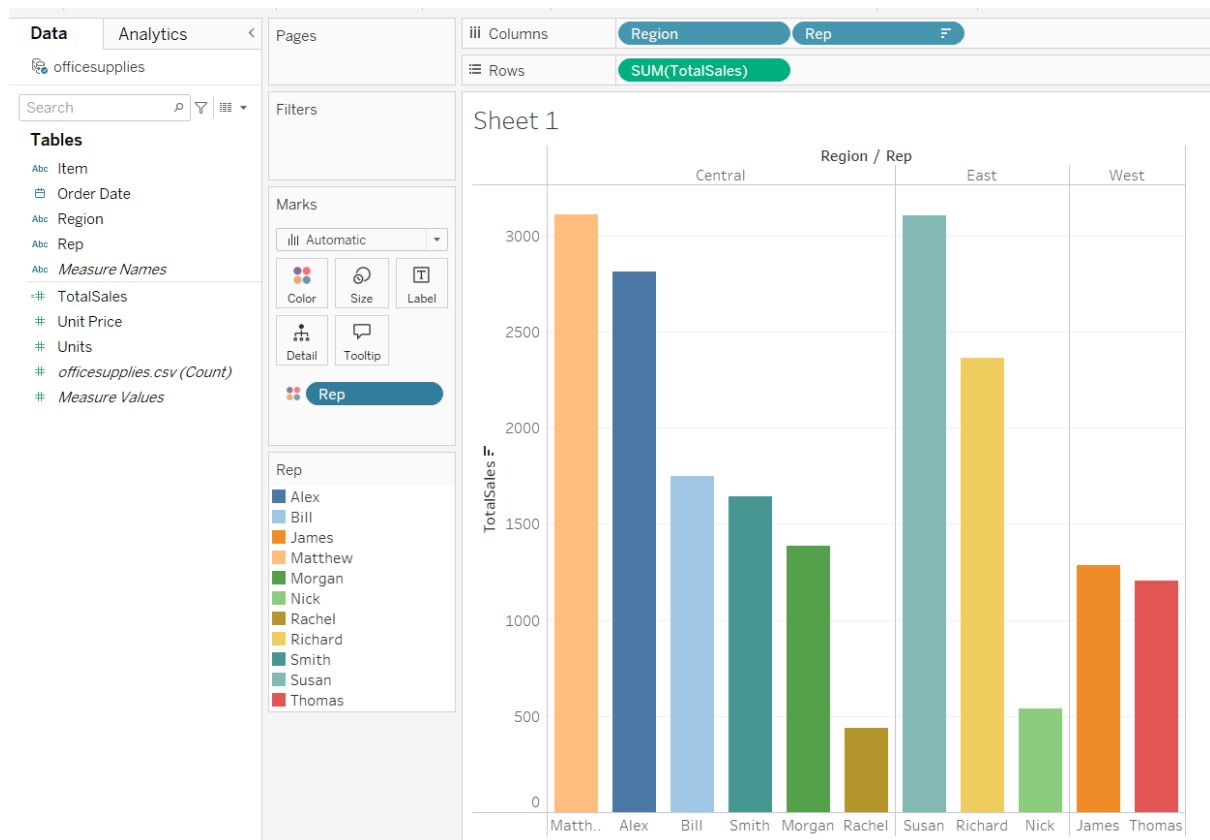
We can change the colors by normally clicking on the Color Mark and directly changing it.



You can take Rep and drag it on the color button.
This will give each representative a unique color.



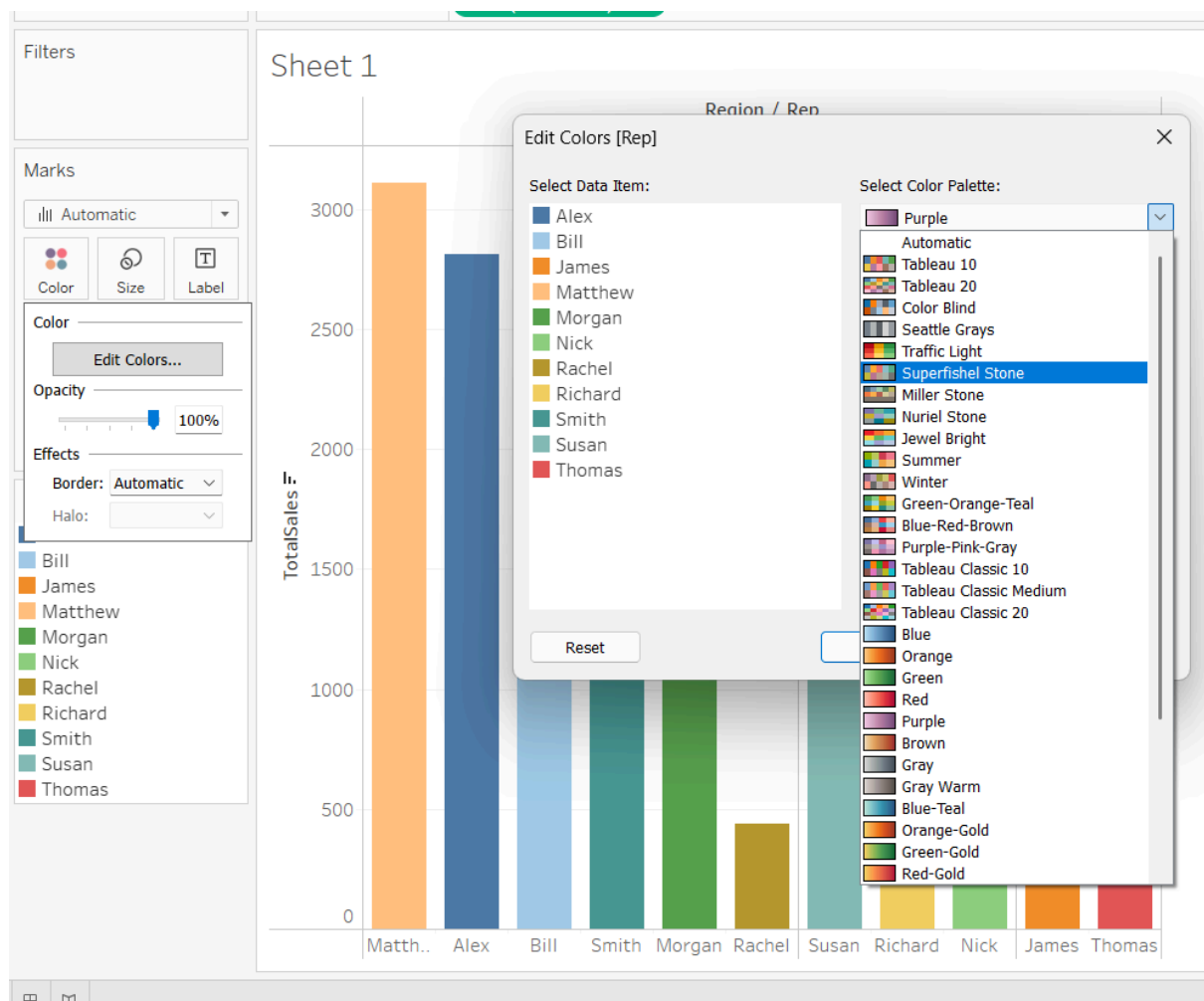
Drag and Drop the color legend on the specified place.



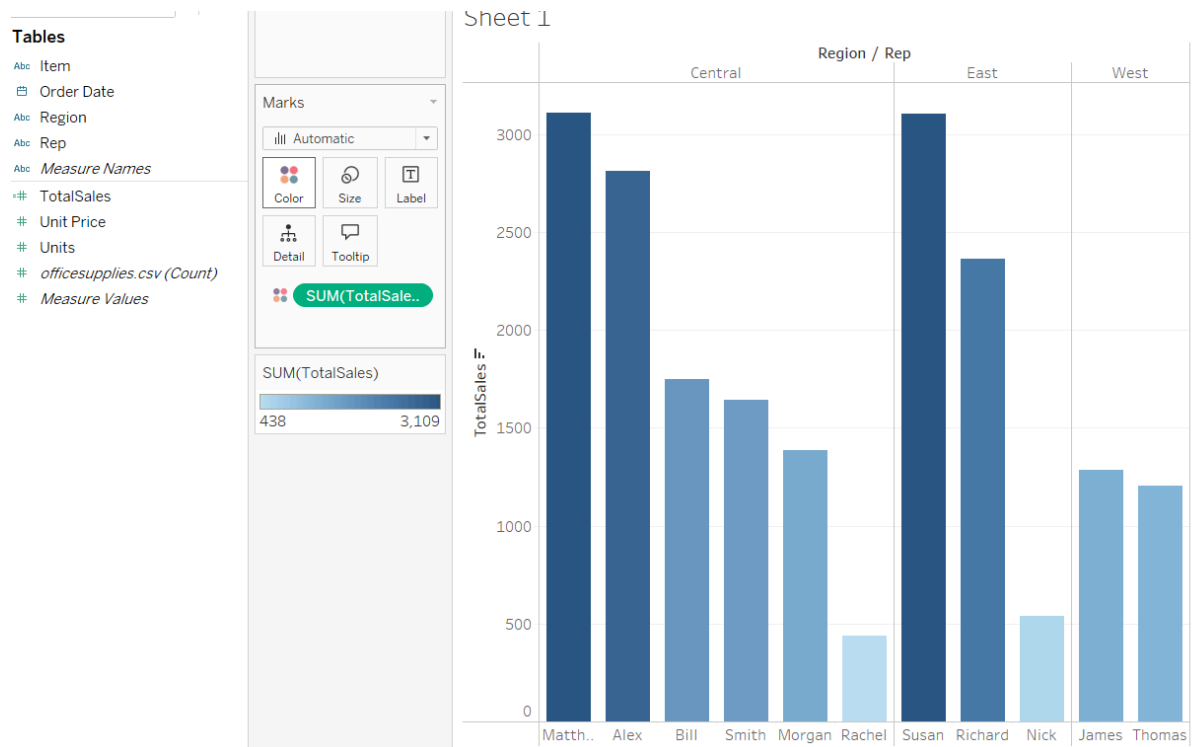
Here the color legend matches the column names.

We can also change the color pallets.

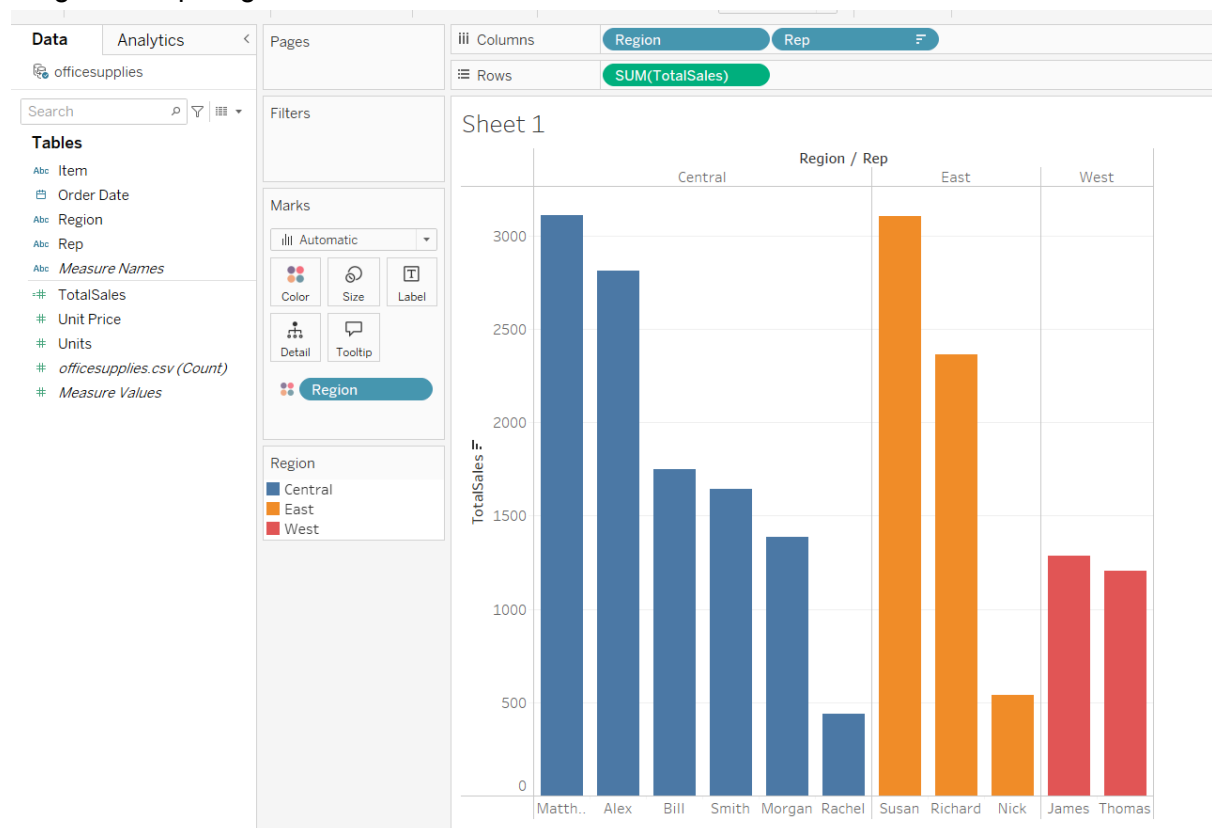
Color → Edit Colors → Color Palette



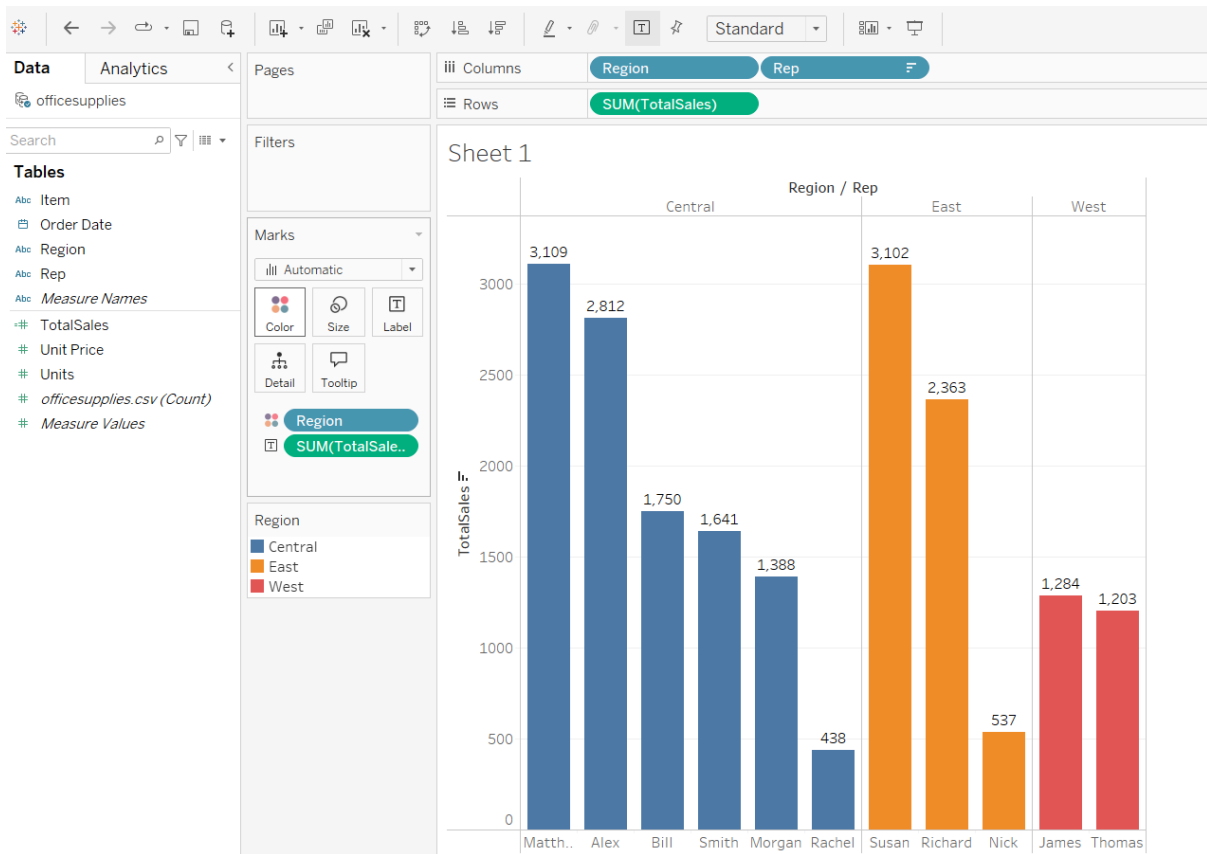
Drag and drop TotalSales in the Color button.



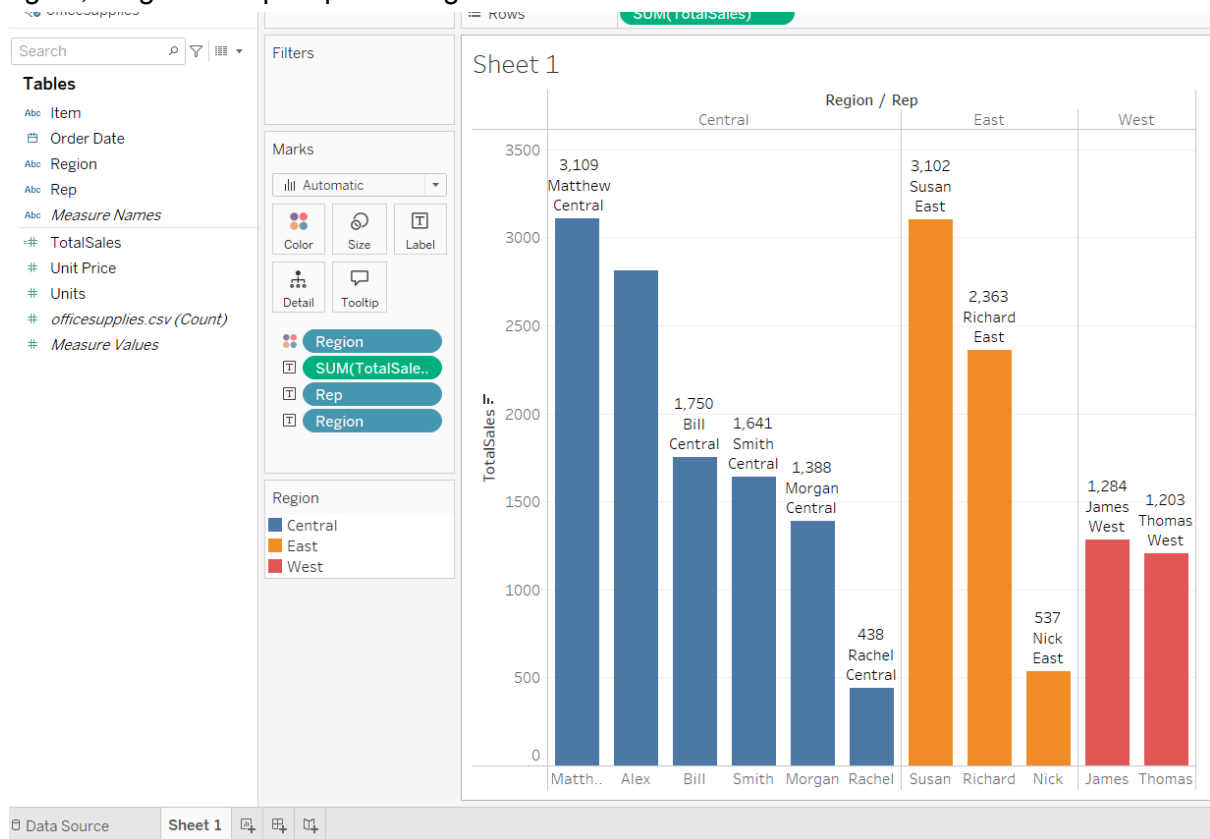
Drag and drop Region on the Color button.



5) Adding Labels and Formatting
Drag and Drop TotalSales on Label.



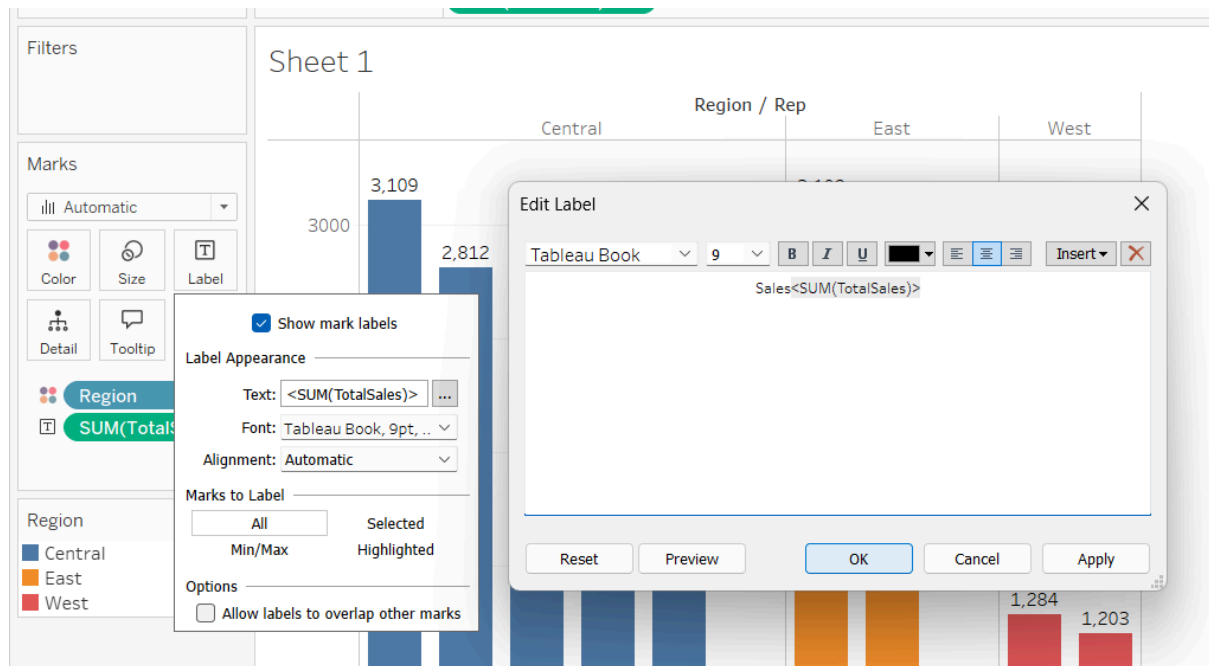
Again, drag and drop Rep and Region on the label.



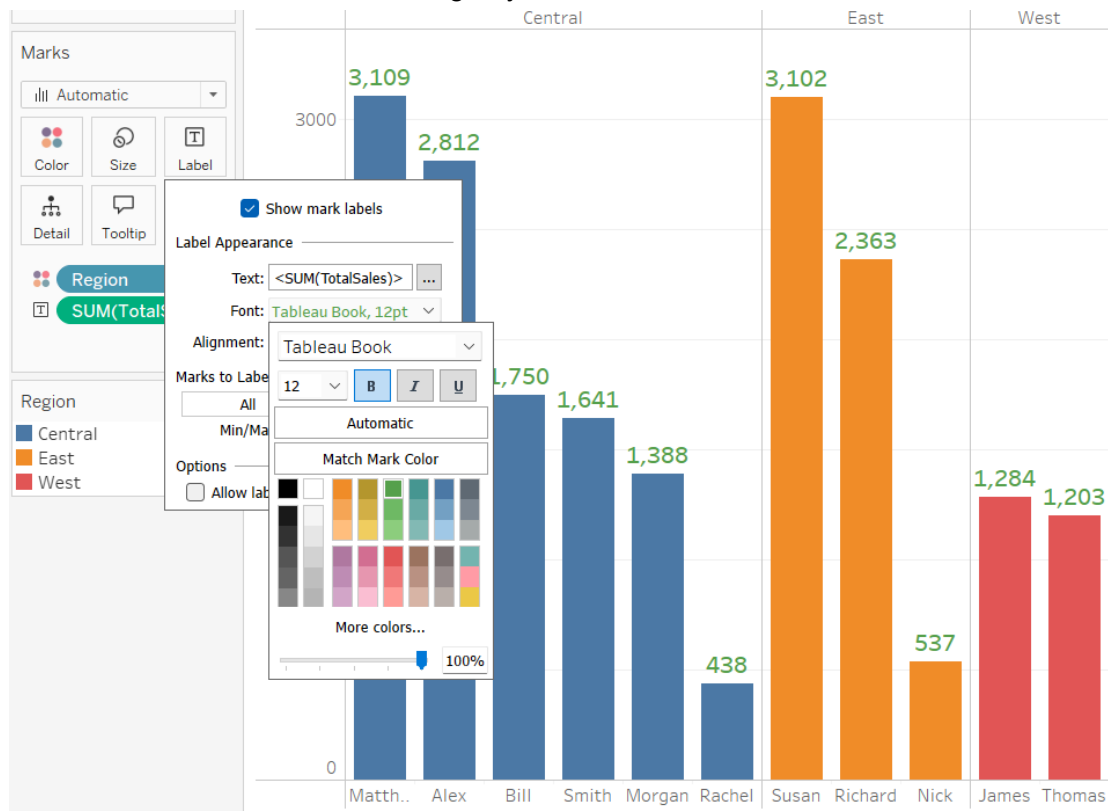
Take Rep and Region out.

We can edit the label further.

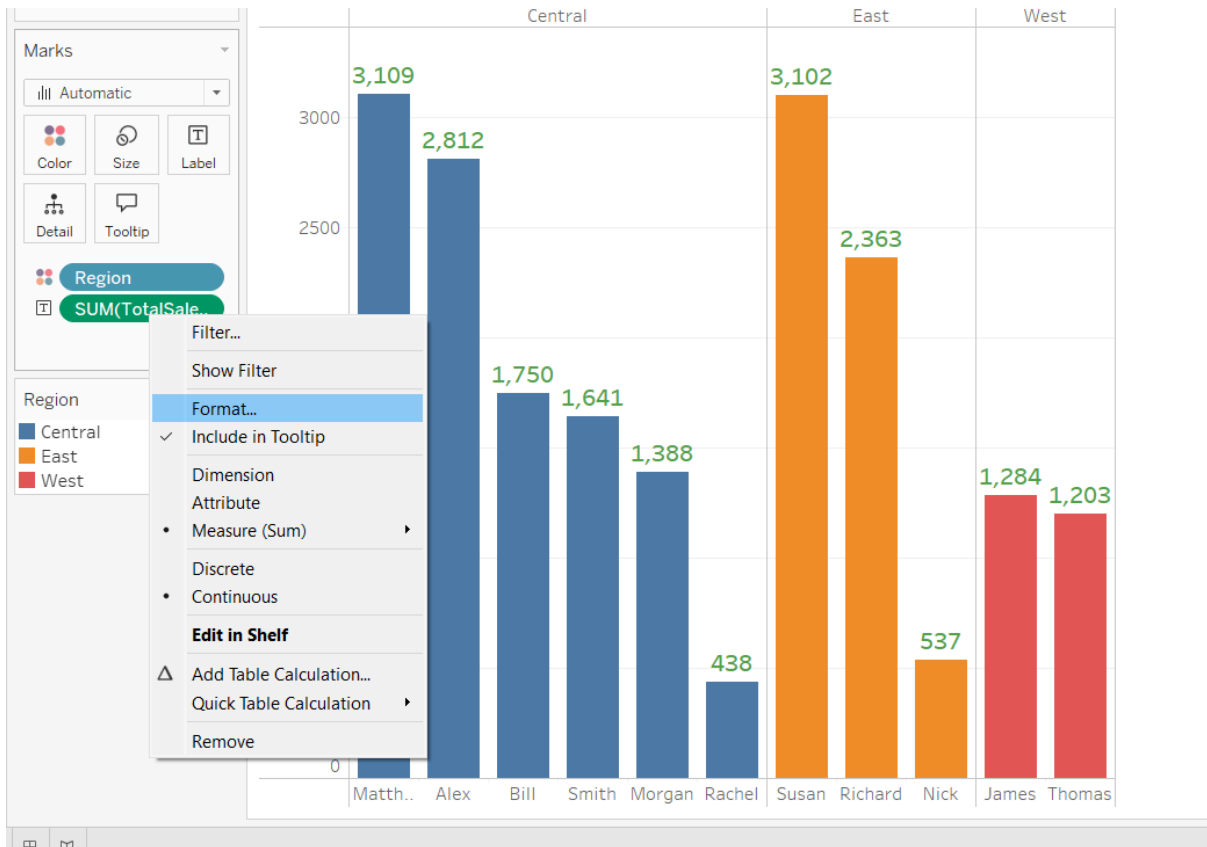
We click the Label button and we can add our own text.



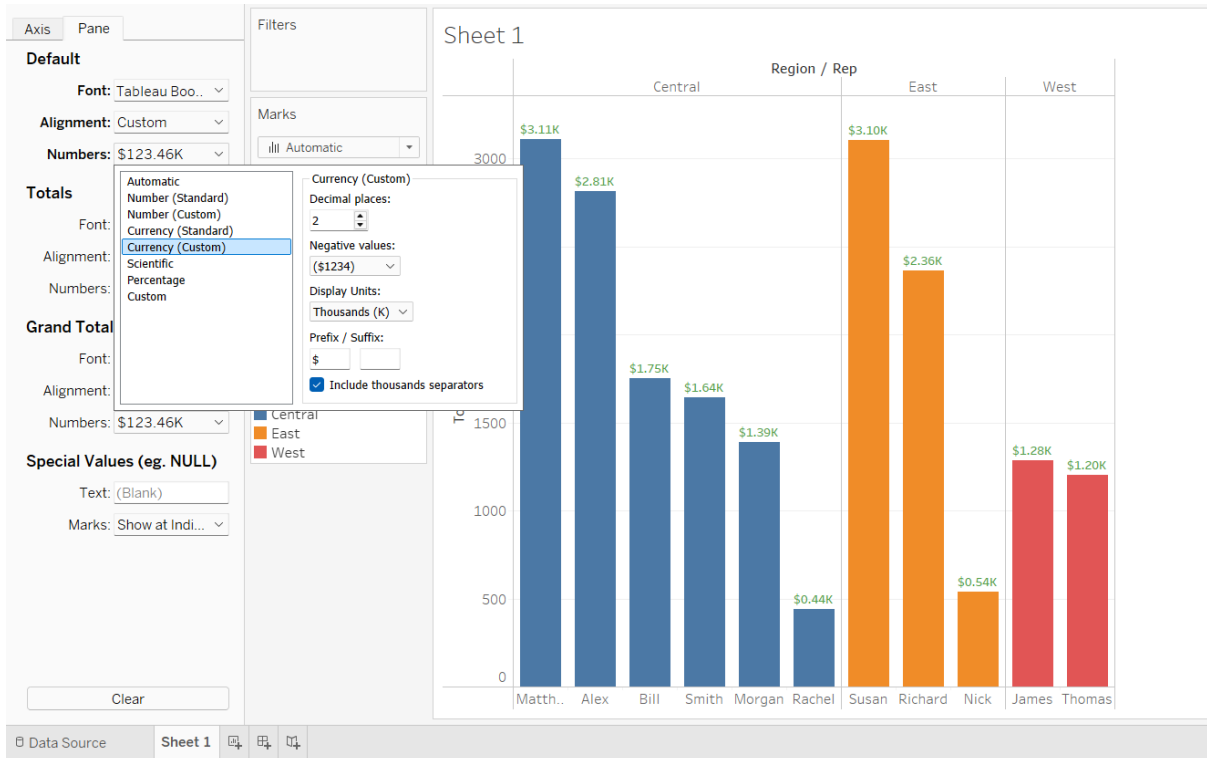
You can format the Label according to your condition.



Right click on SUM(TotalSales) and Click on Format.

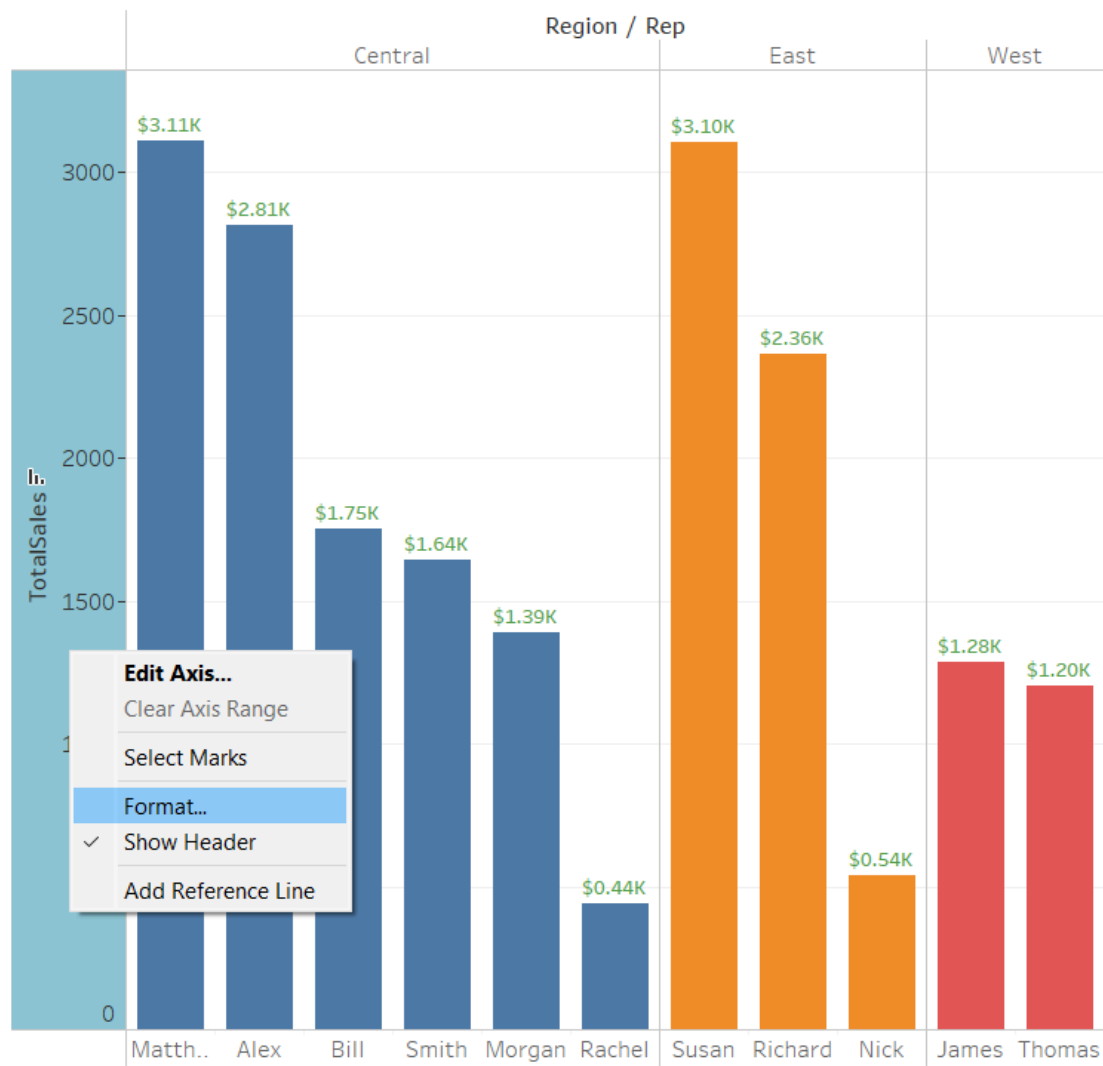


Display this as \$ and K values.

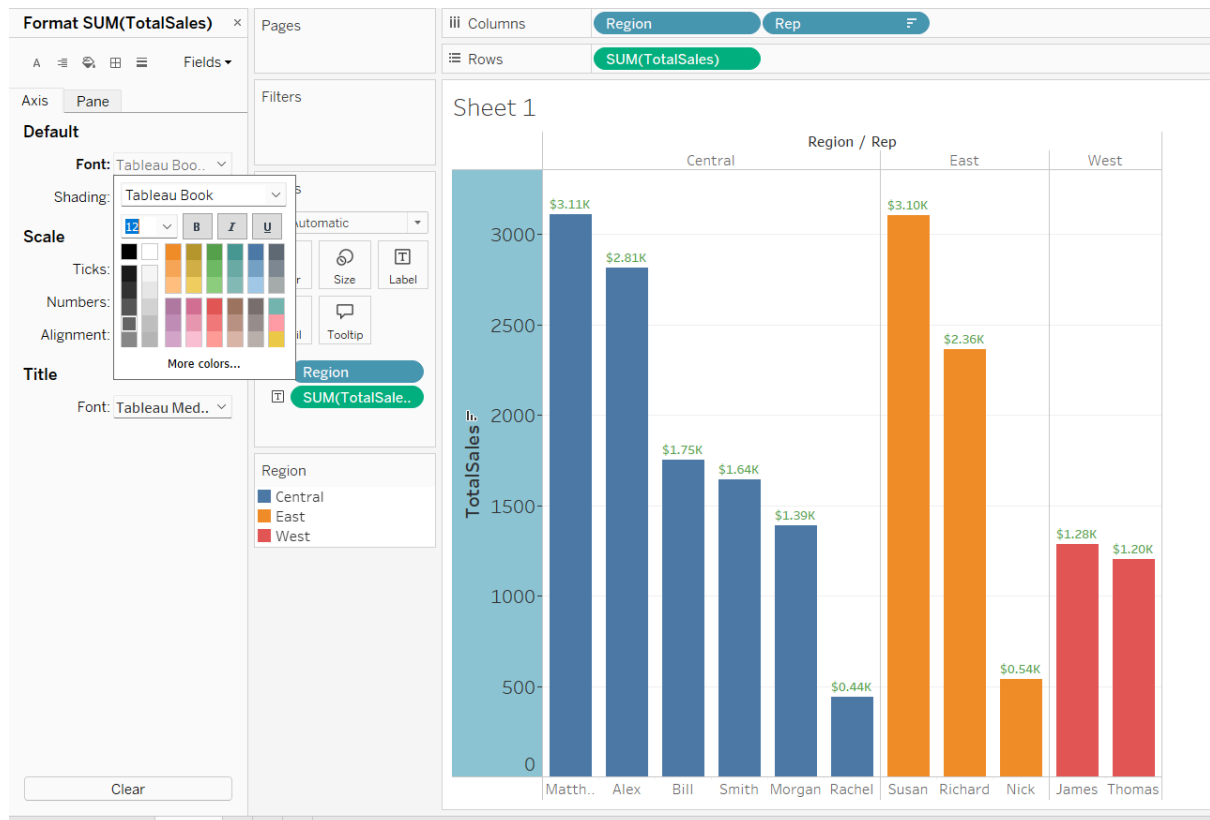


Right click on the Axes and click on Format.

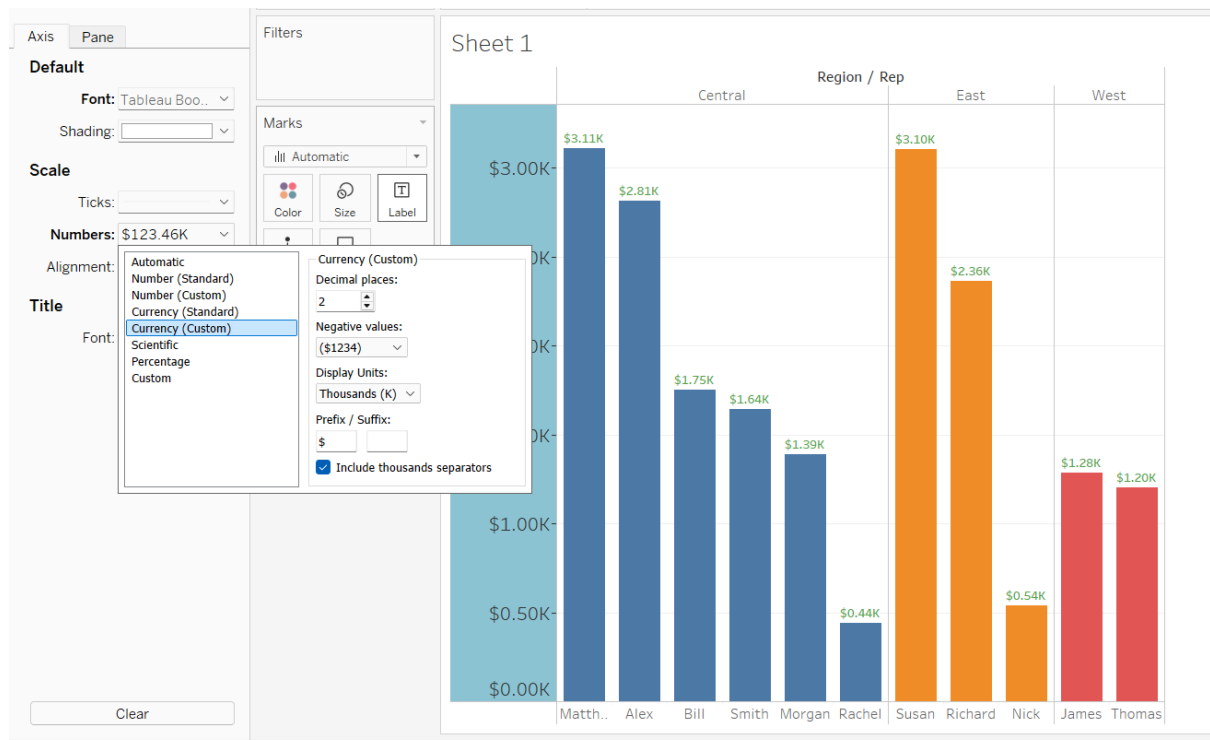
Sheet 1



Change the font size.



Make the axes of TotalSales also in \$ and Display Units in K.



17) Export the Worksheet

Time Series Aggregation and Filters

Import the Unemployment data.

1) Working with Time Series data

Right click on Data source and then click on View Data.

View Data: Sheet1 (Unemployment) ×					
Tables	Sheet1 1,708 rows 4 fields Show Fields				
Sheet1	Age	Gender	Period	Unemployed	
	16 to 19 years	Men	1/1/2005	91,000	
	20 to 24 years	Men	1/1/2005	175,000	
	25 to 34 years	Men	1/1/2005	194,000	
	35 to 44 years	Men	1/1/2005	201,000	
	45 to 54 years	Men	1/1/2005	207,000	
	55 to 64 years	Men	1/1/2005	101,000	
	65 years and over	Men	1/1/2005	33,000	
	16 to 19 years	Women	1/1/2005	38,000	
	20 to 24 years	Women	1/1/2005	90,000	
	25 to 34 years	Women	1/1/2005	142,000	
	35 to 44 years	Women	1/1/2005	180,000	

Double click on Unemployed and double click on Period.

☰ Rows	SUM(Unemployed)
--------	-----------------

Sheet 1

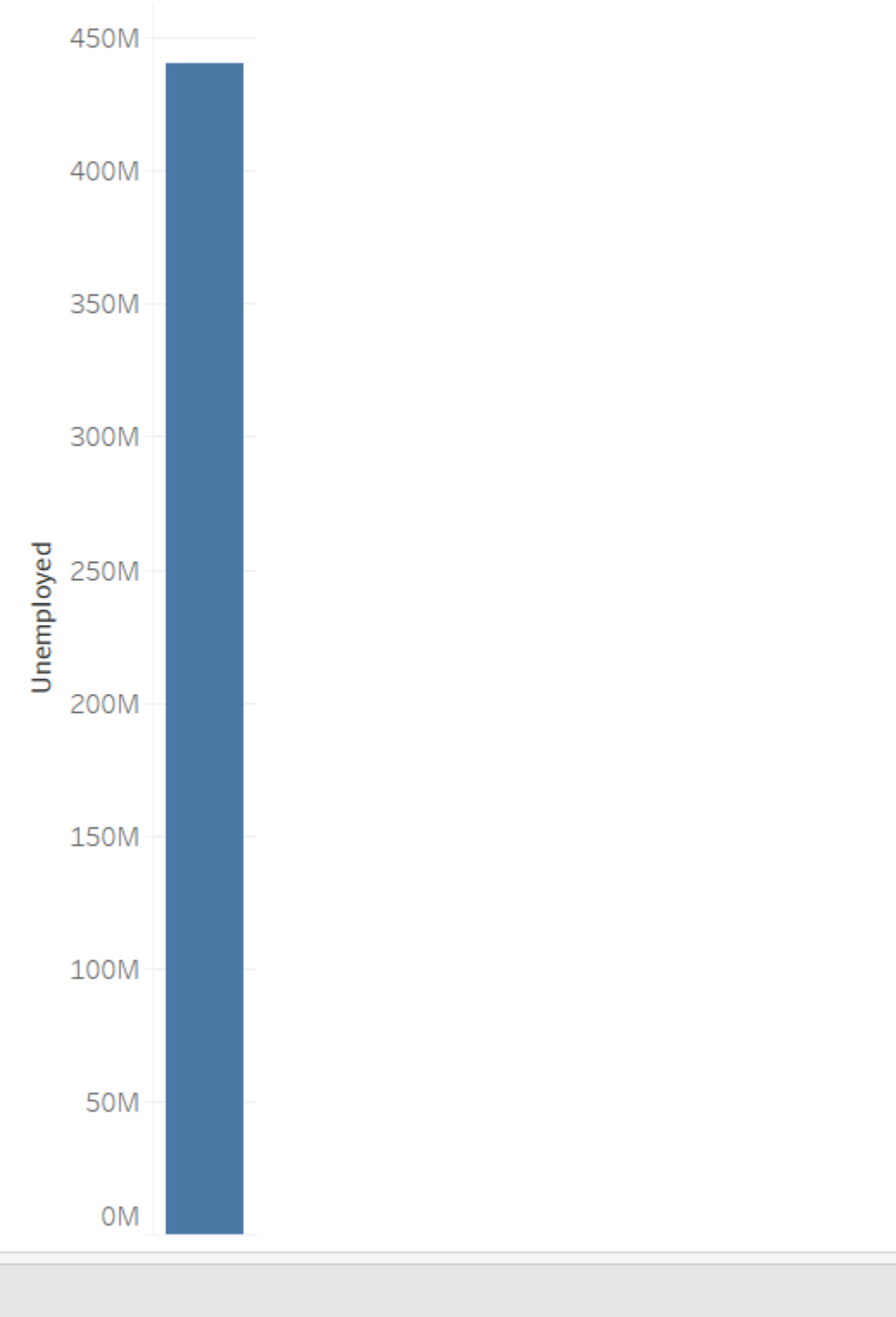
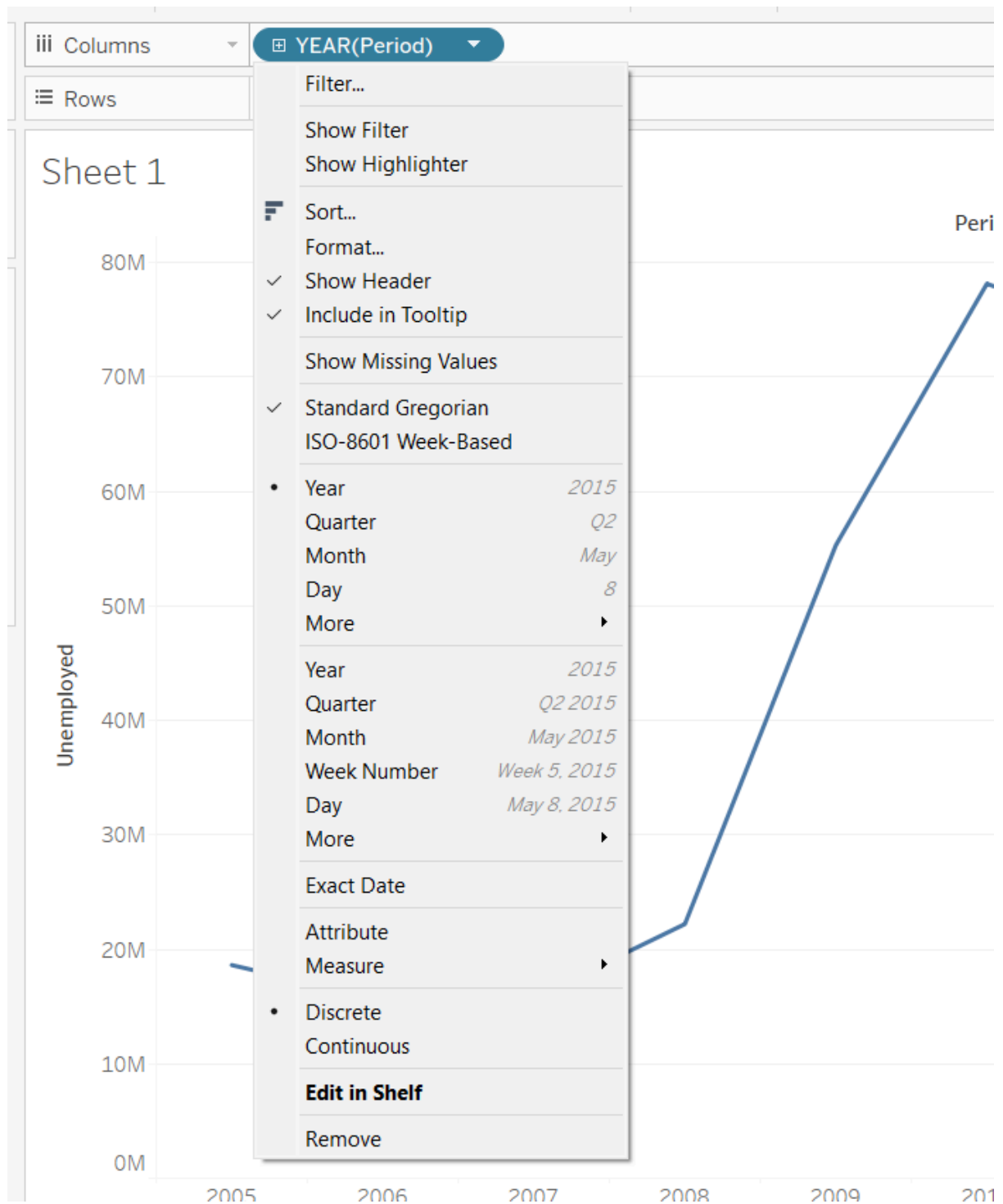




Tableau always highlights dimensions in blue and measures in green.

Go to the dropdown option of YEAR(Period).



There are two sections which are responsible for time.

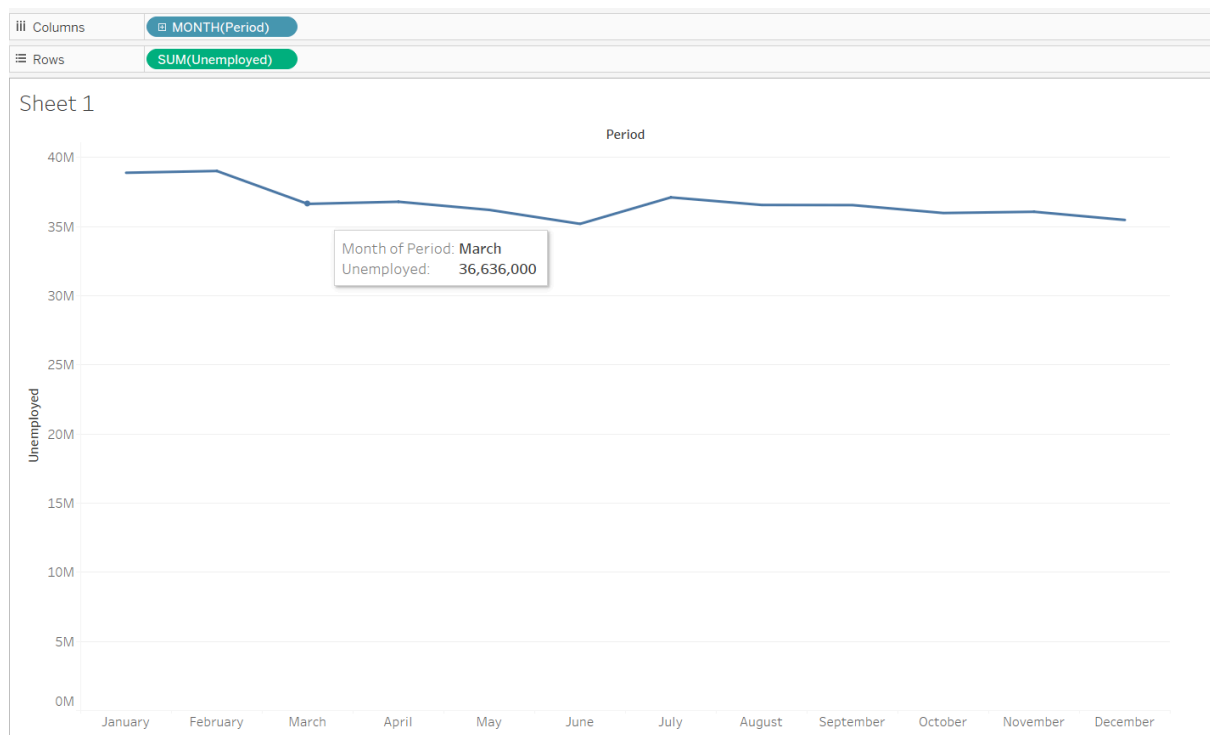
Click on Month of the first section.

We no longer have years on the bottom.

Instead of that we have months.

Tableau is taking all the data of the months irrespective of the year.

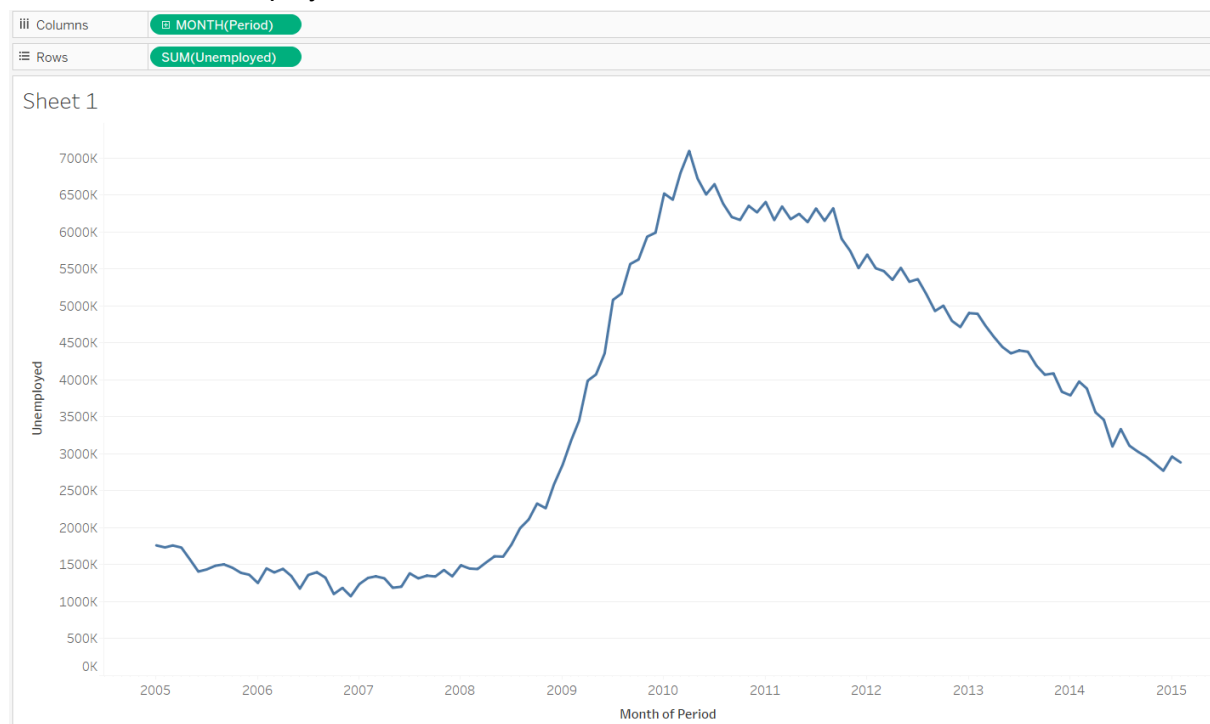
It will take and aggregate all the values.



Here months are treated as categories instead of timeline.

Select the second month from the dropdown.

Then the chart is displayed as:



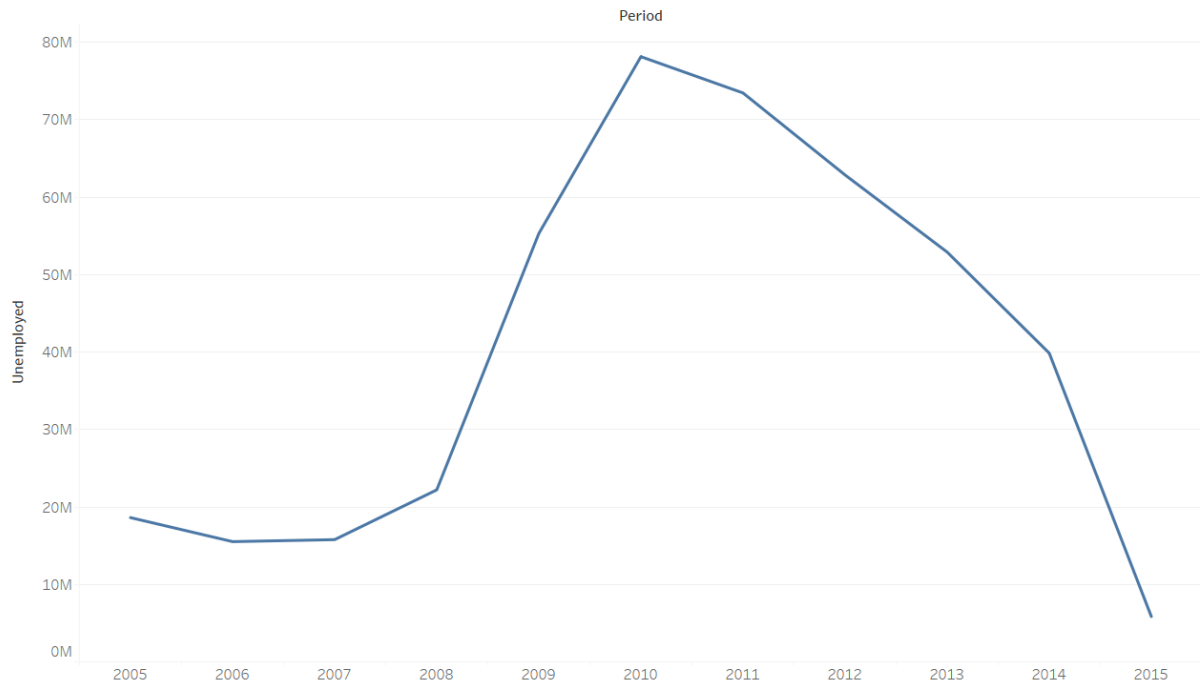
Here the month of period is also green.

So, it is recognized as a measure.

Select Year also from both types.

First year gives:

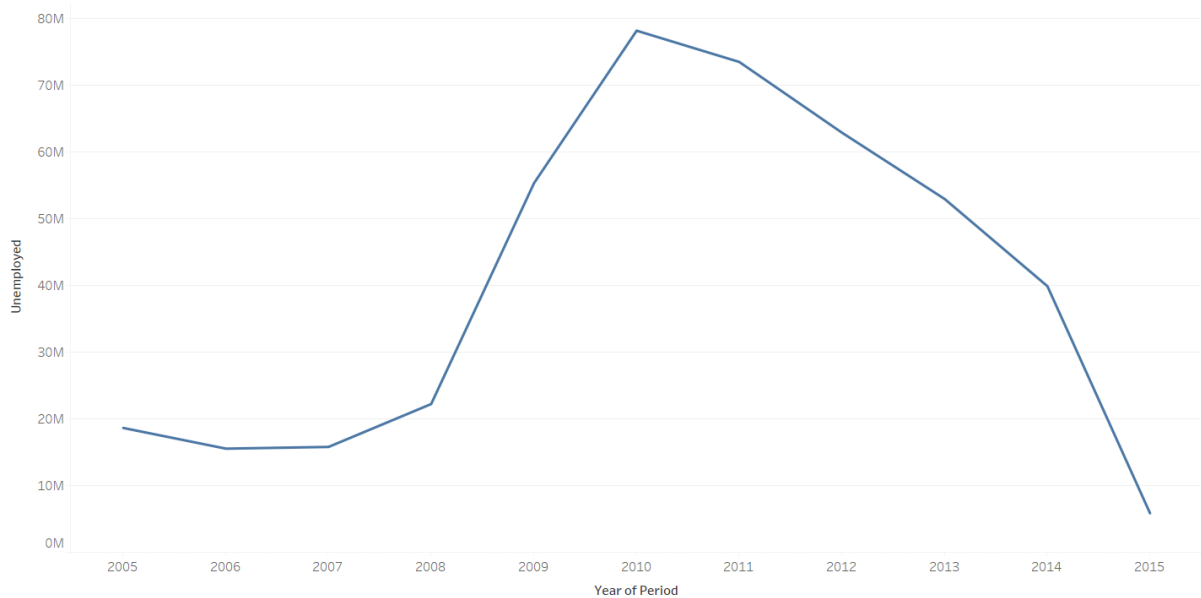
Sheet 1



Here we got categories of the year.

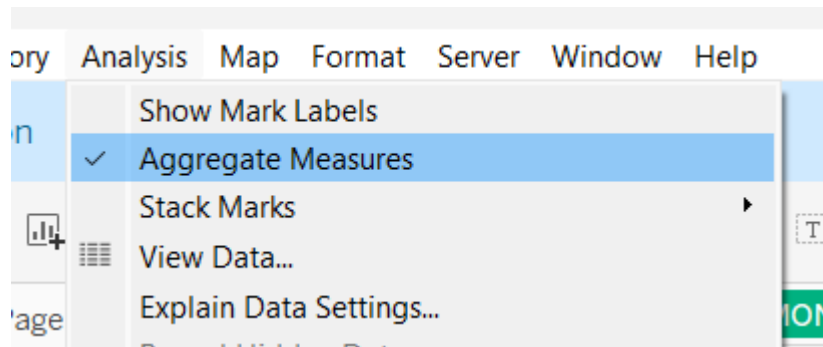
Second year gives:

Sheet 1

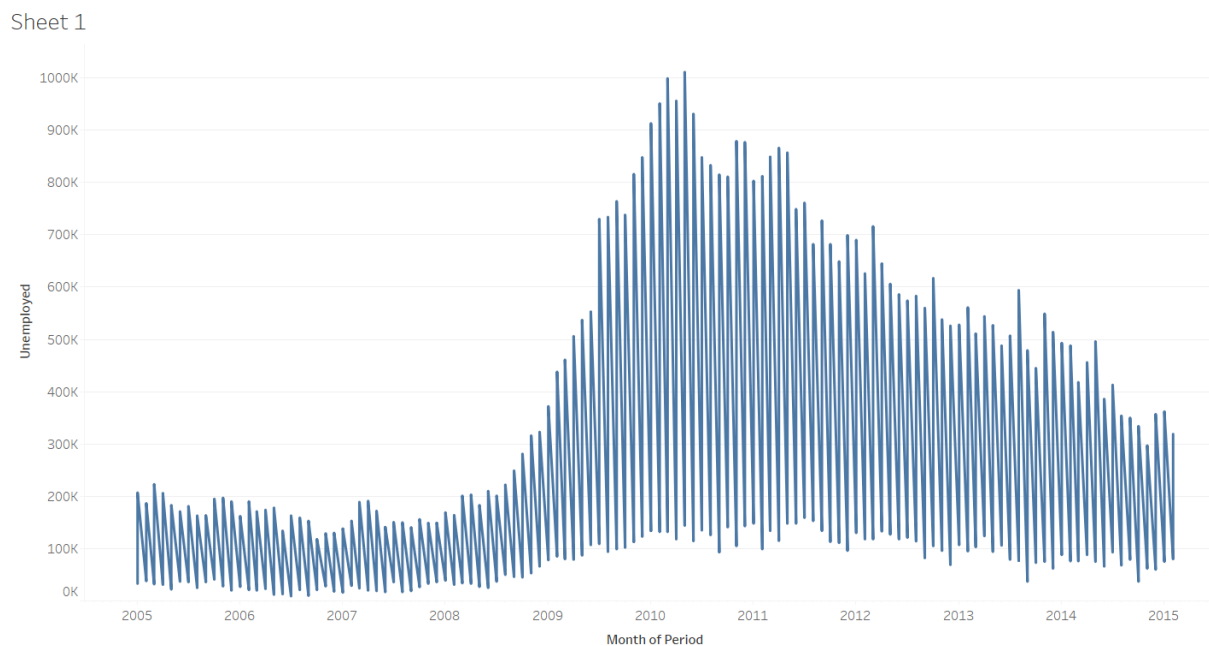


Here we got the timeline of the year.

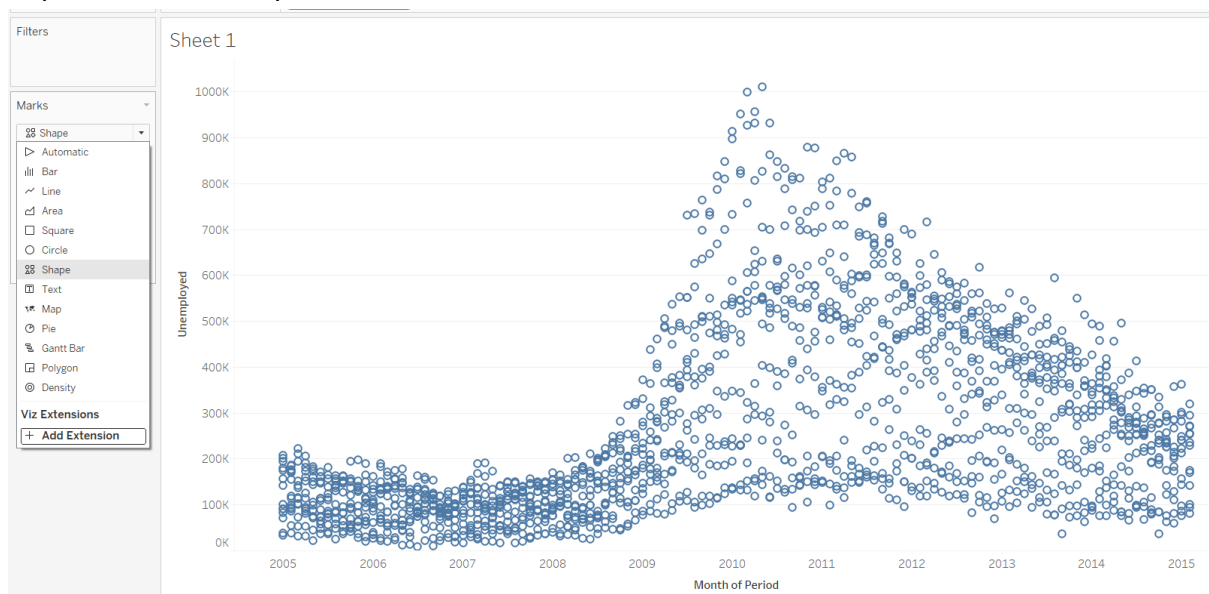
Plot every single row as a dataset.
Go to Analysis and uncheck Aggregate Measures.



We will get this type of chart.

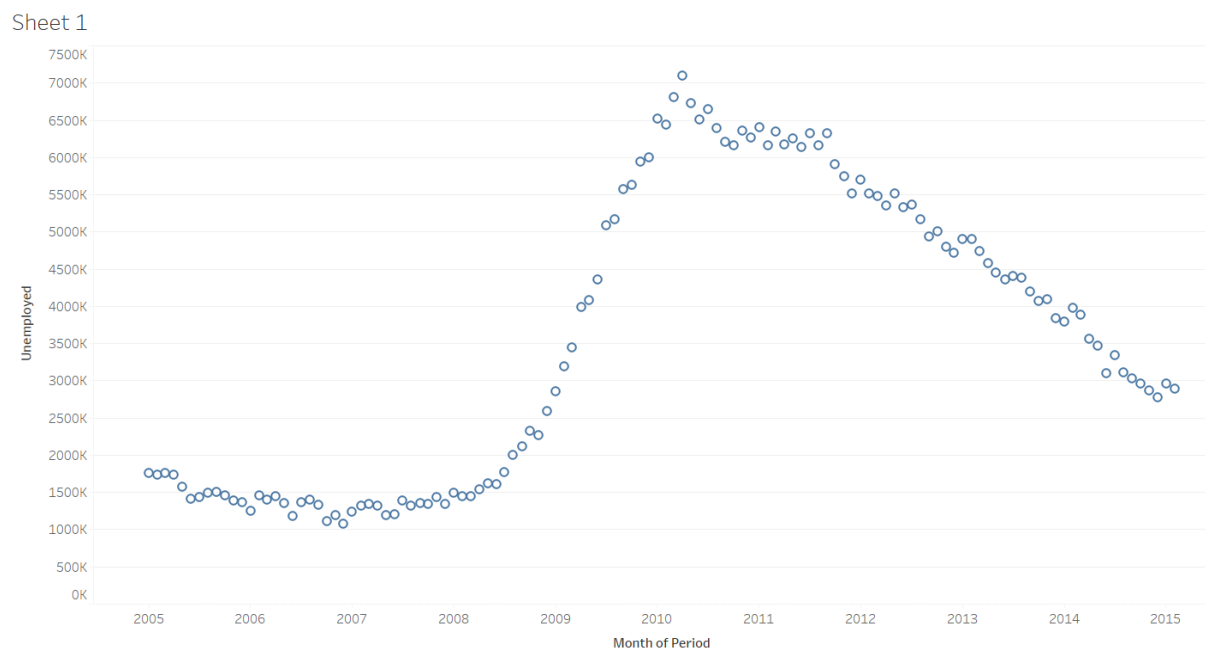


Replace it with a shape chart.



Here tableau is plotting every single row of our dataset separately on this chart.

Now go back and check Aggregate Measures.



Drag and Drop Gender to the Color.

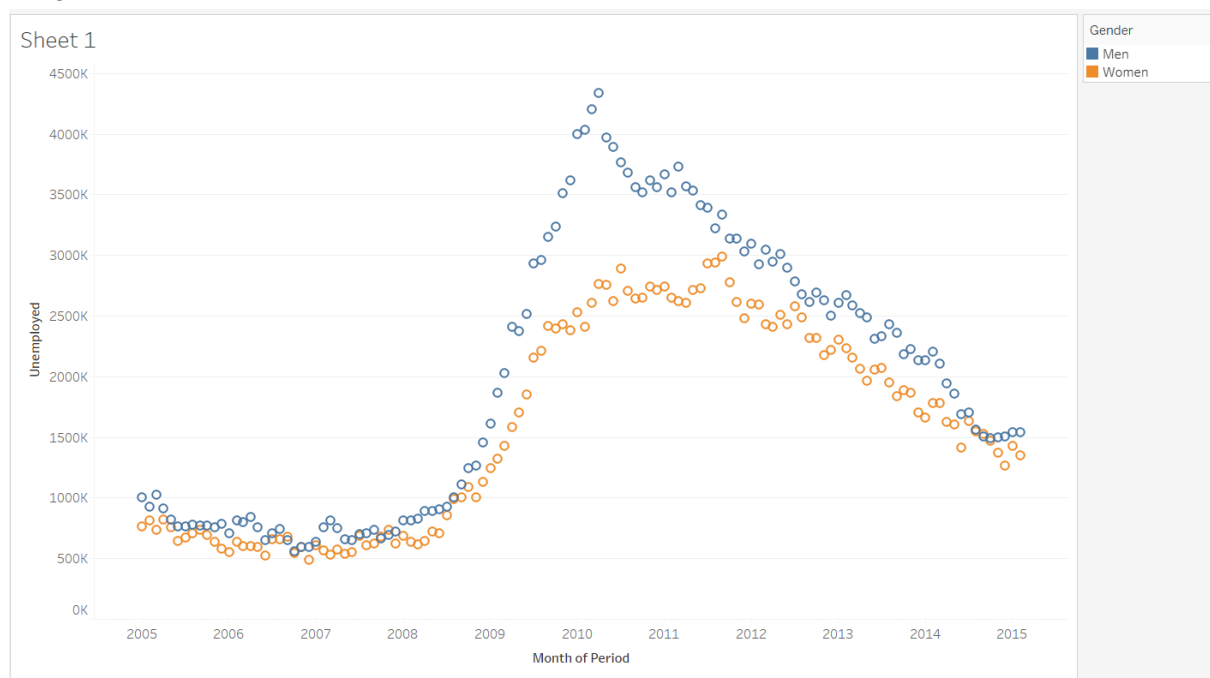
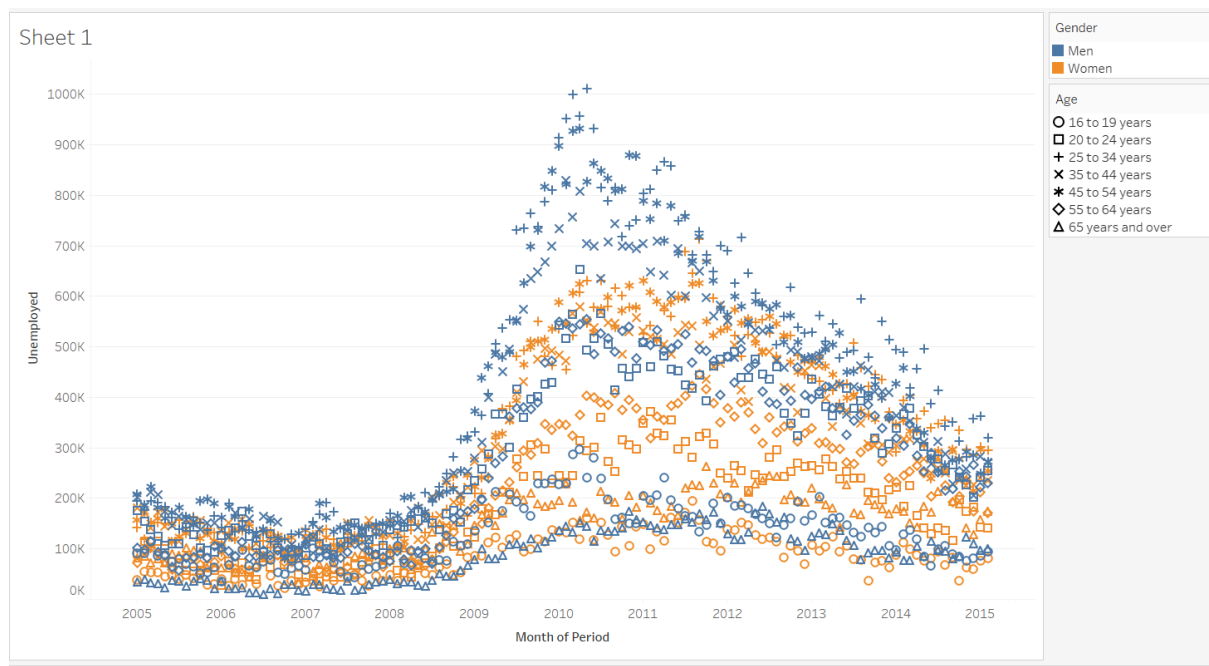


Tableau automatically plots the gender data.

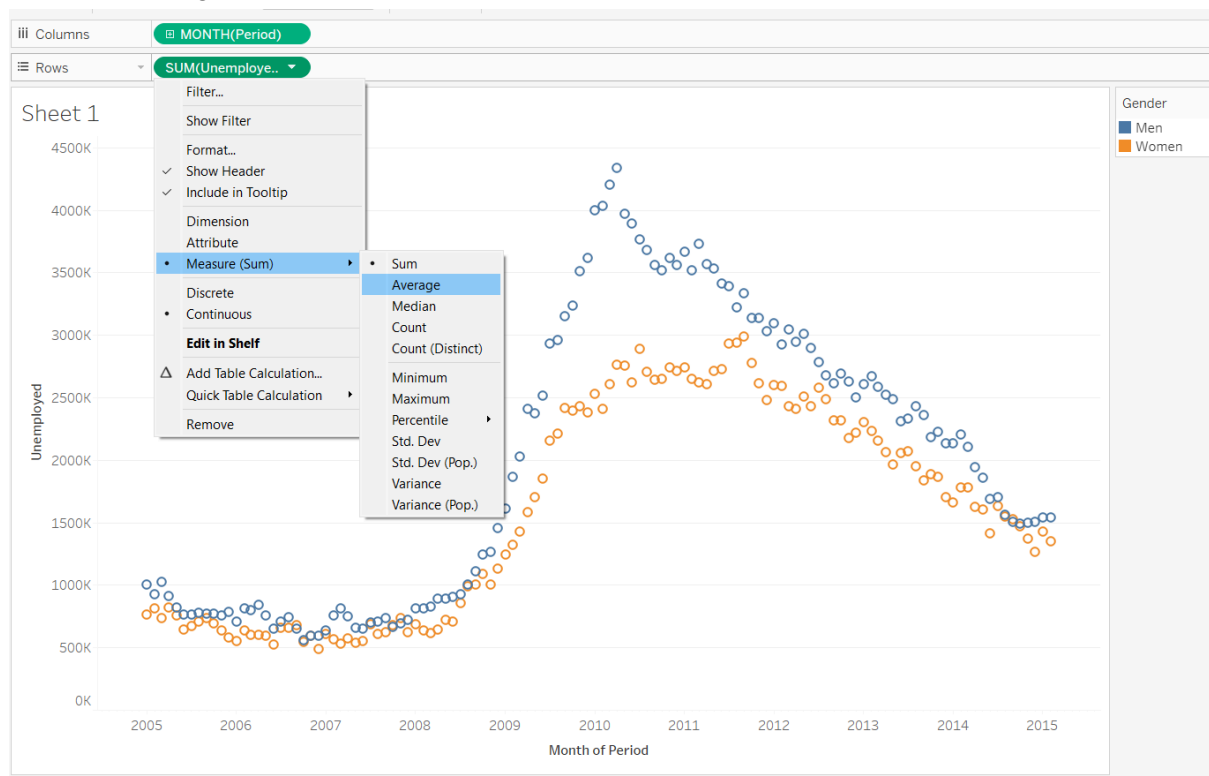
Blue dots represent male and orange dots represent female unemployment for that specific month.

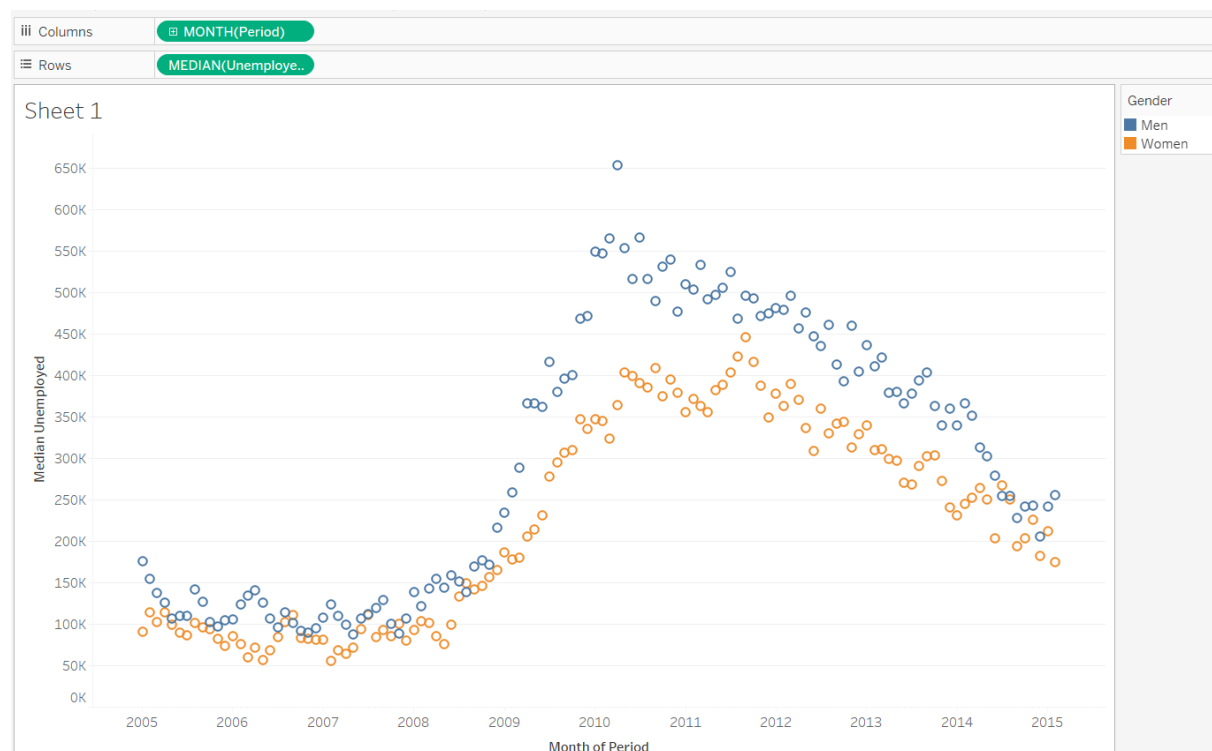
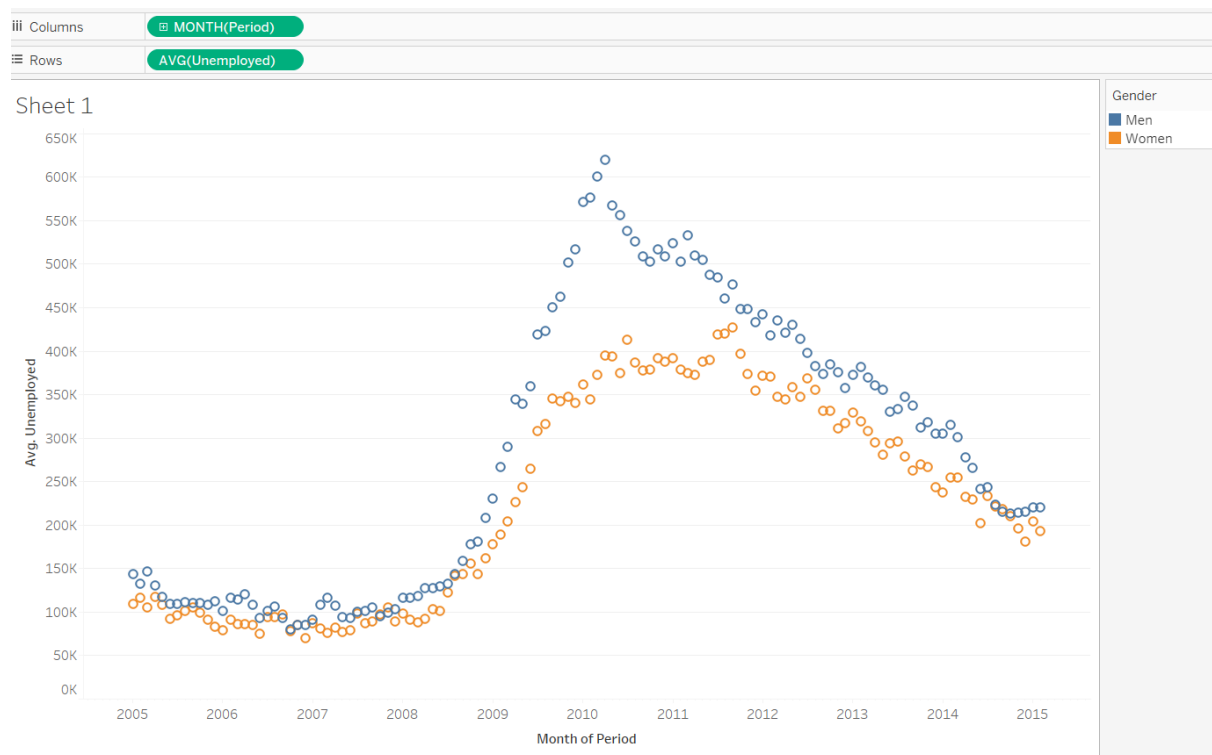
Drag and Drop Age to Shape.



Remove age.

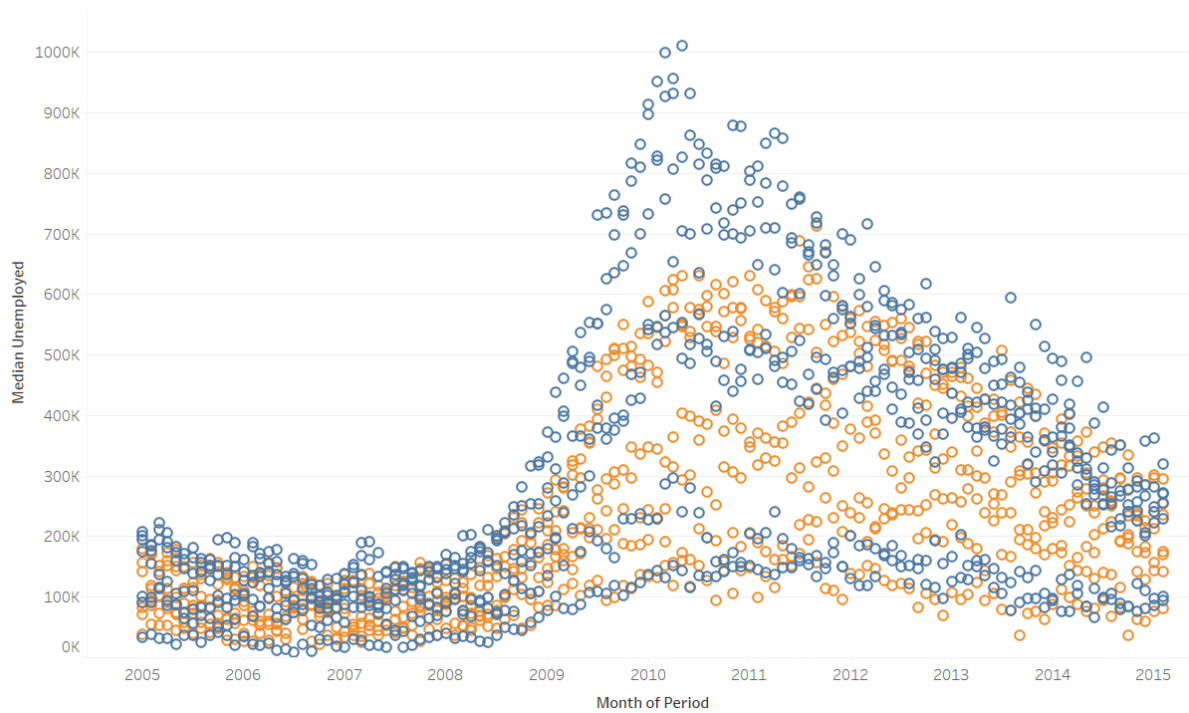
Let's find the age instead of the sum.





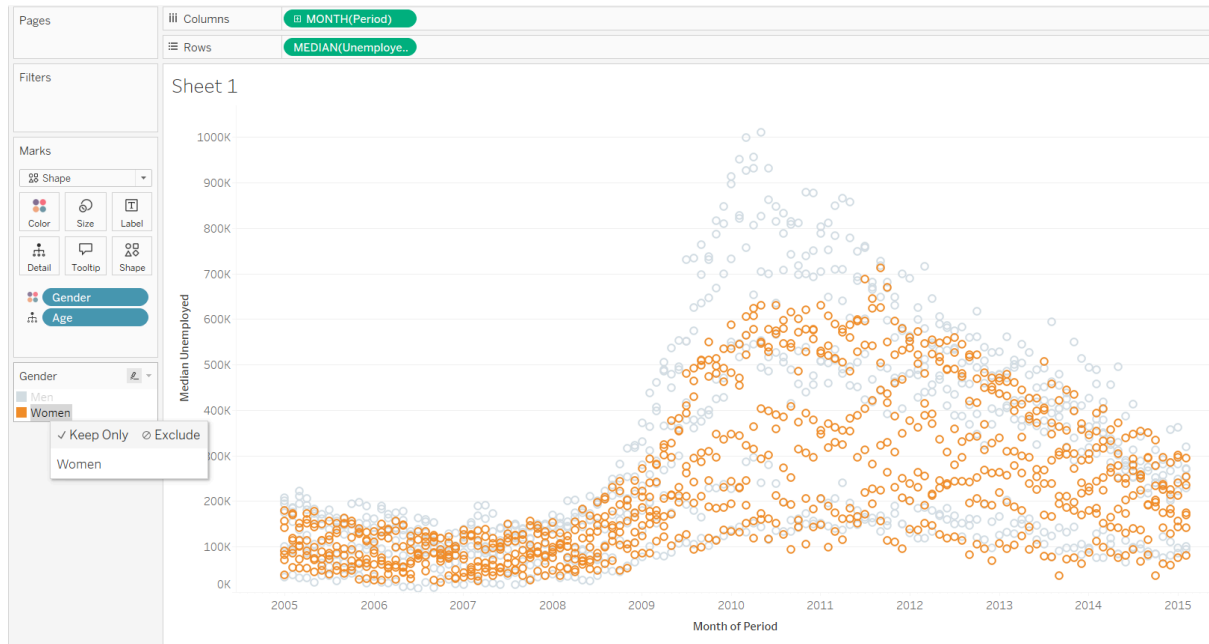
Make the dashboard more detailed without affecting the visual.
Drag Age and put it on Detail.

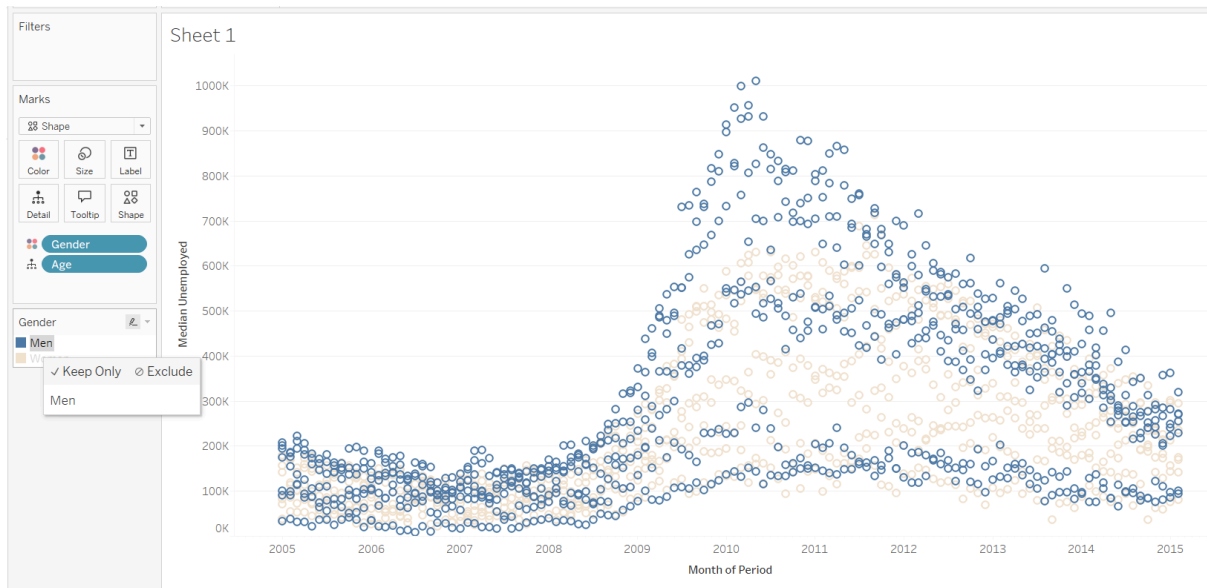
Sheet 1



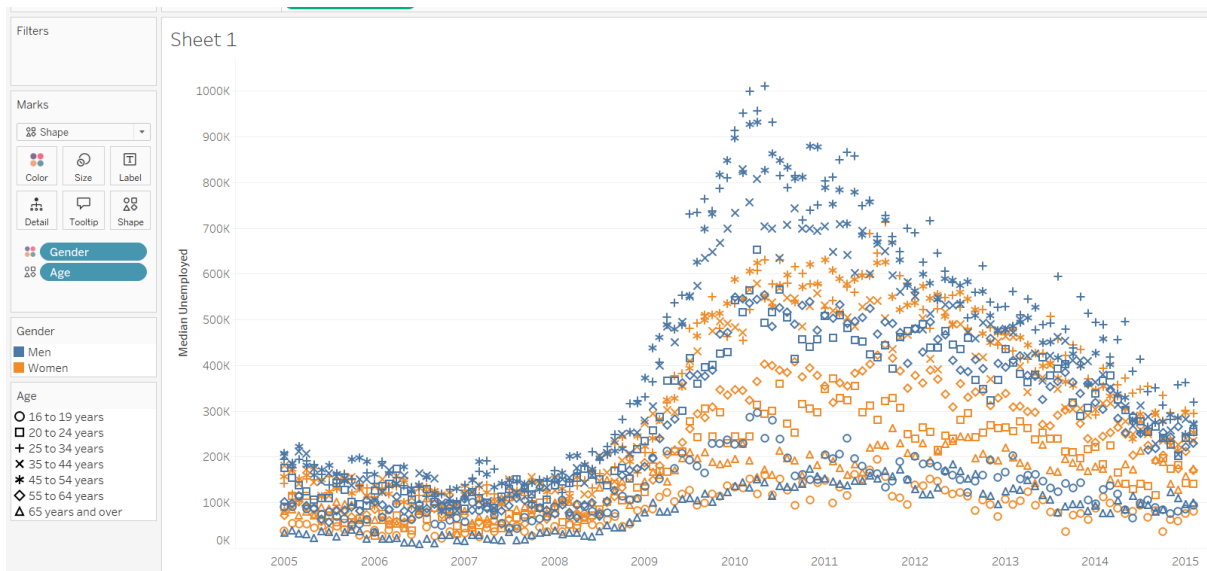
Highlighting.

Click on the label to highlight the specific data.

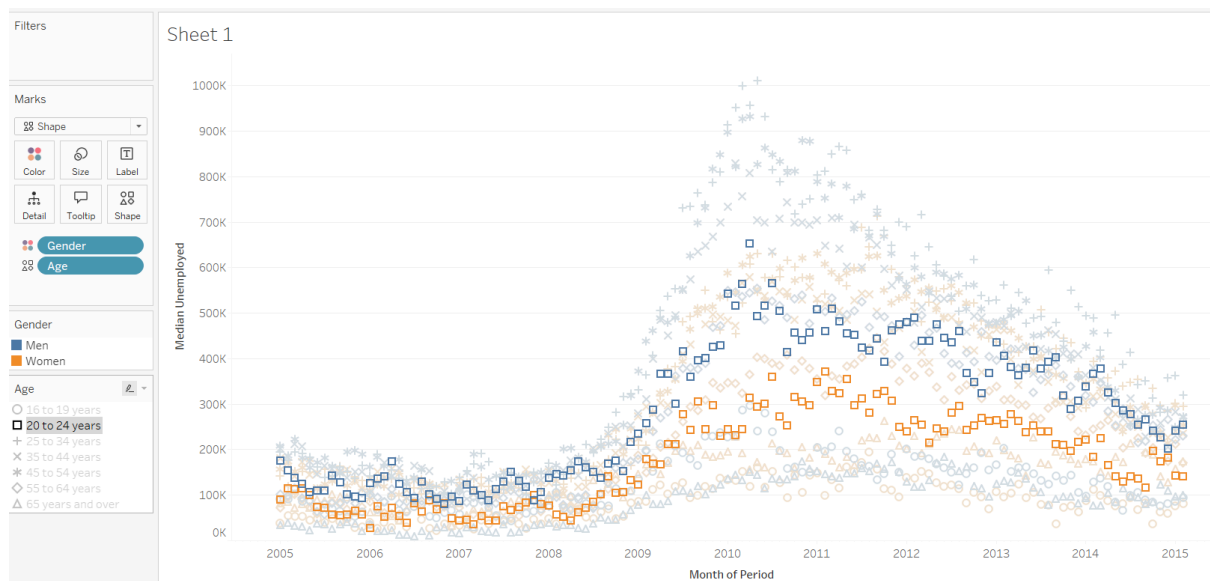




Drag and drop Age to Shape.



Click on specific Age range and click on the Highlight Selected Items button.



Move back to our original line chart.

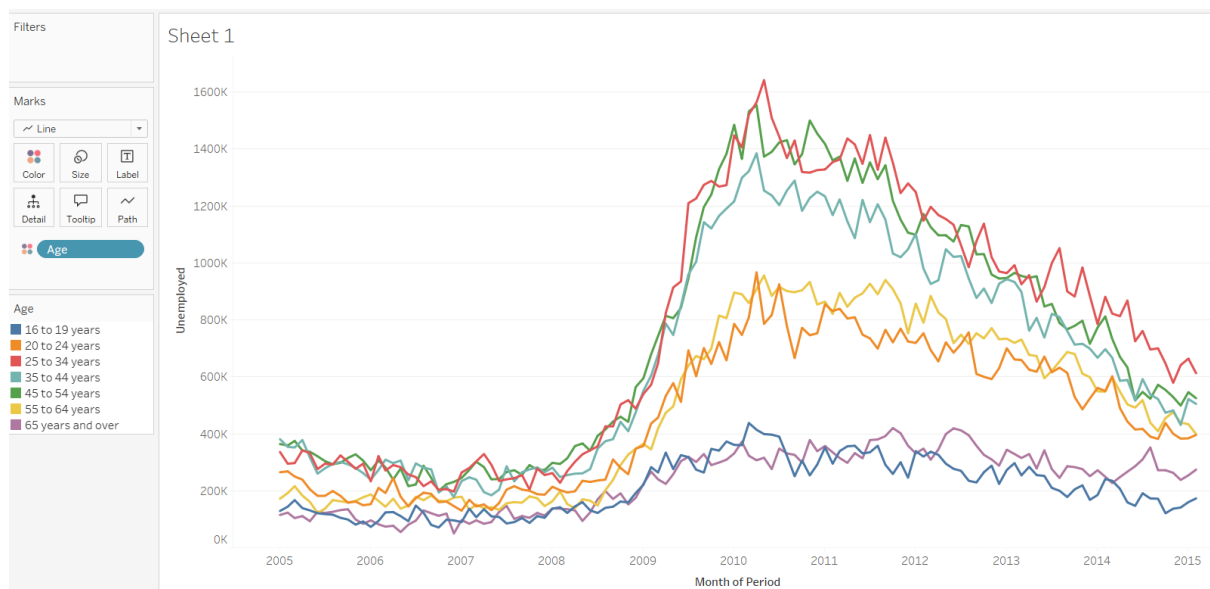
Take Age and Gender out.

Change Shape to Line.

Change Median to Sum.

Take Age and drag it to Color.

It will give us many line charts.

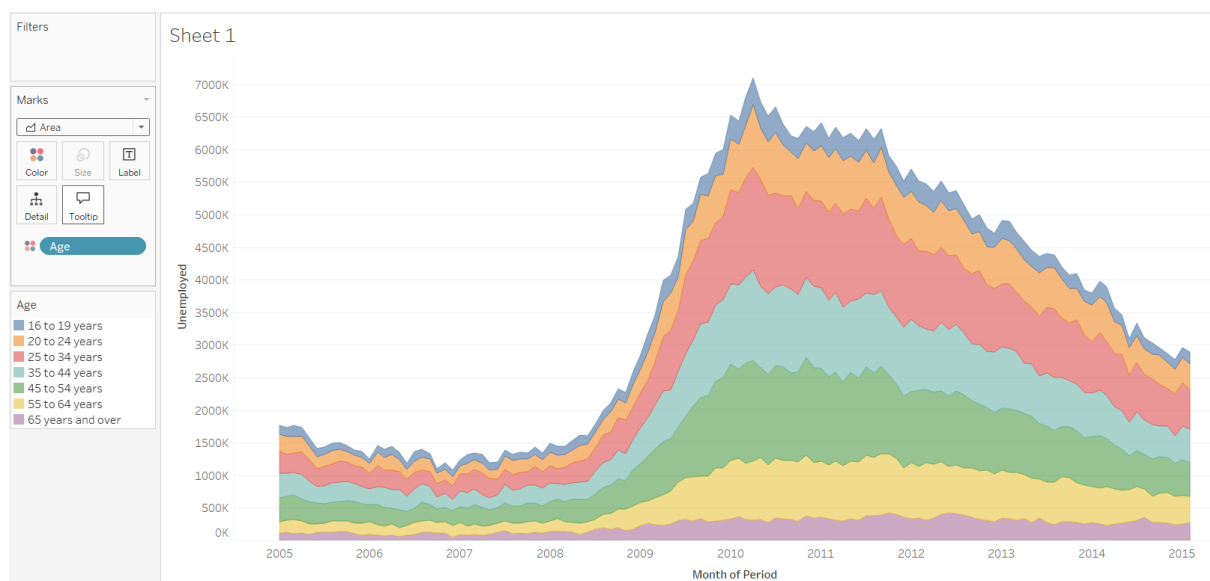
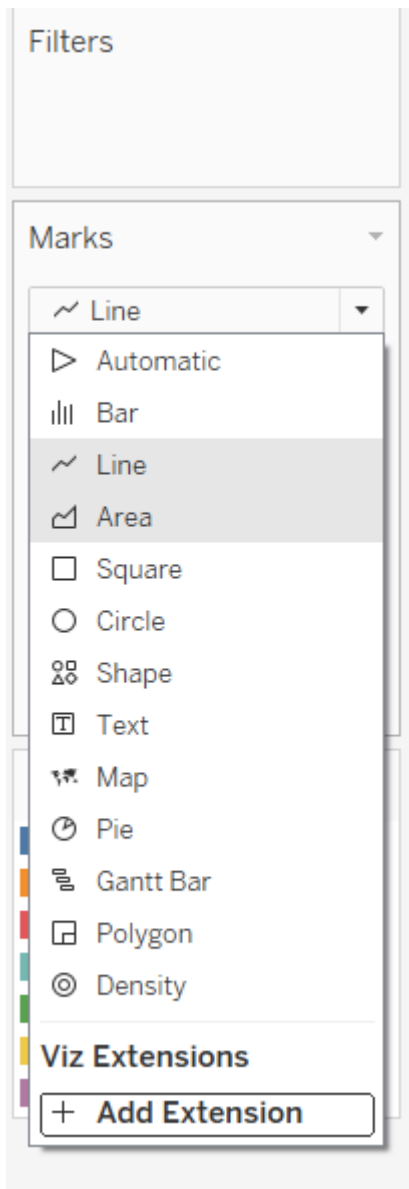


This is not easy to understand.

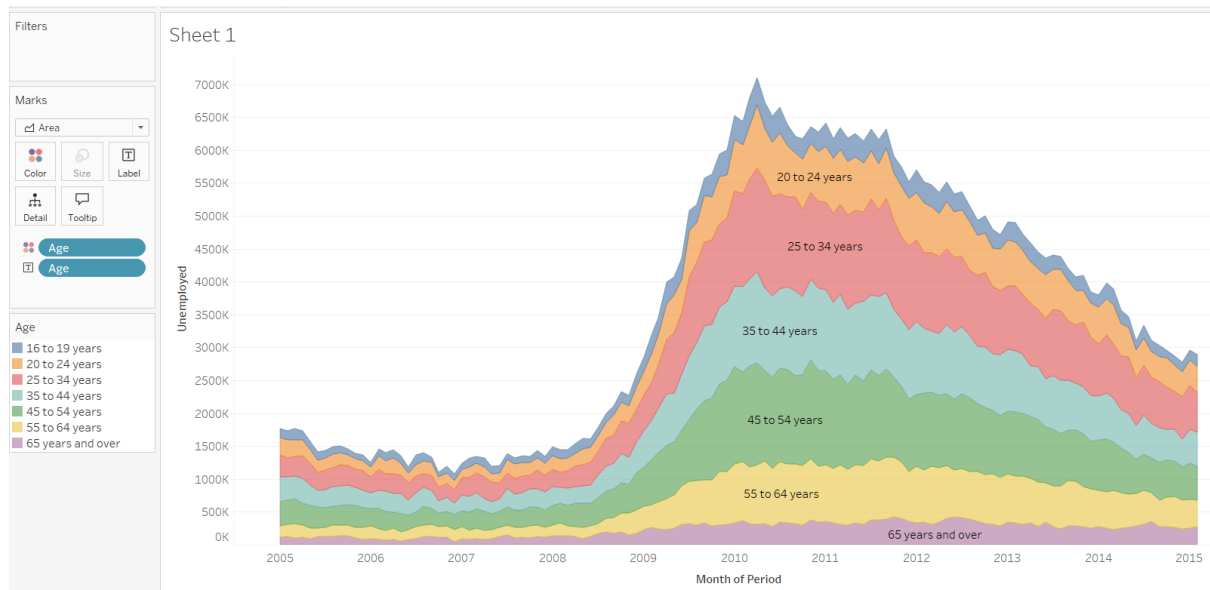
So change the visualization to a different type.

Try out different types of visualization and finalize the area chart.

Go to Dropdown and Select Area button.

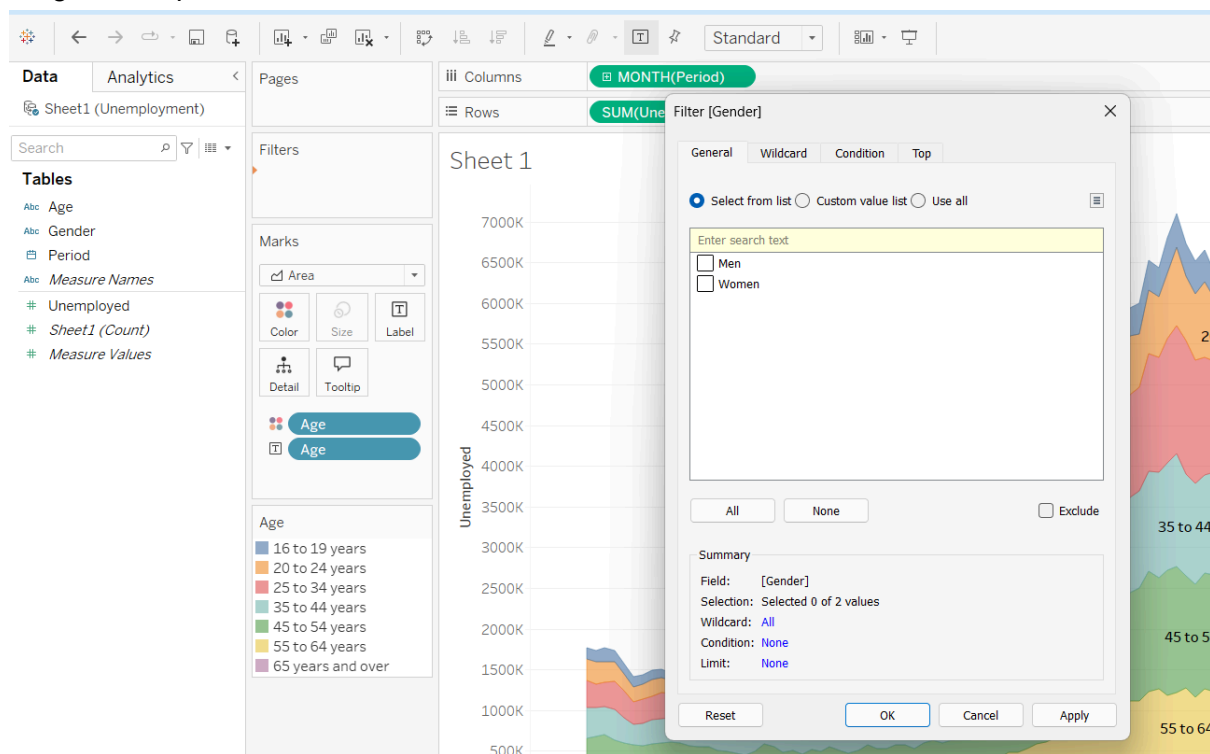


Drag and drop Age to the Label.

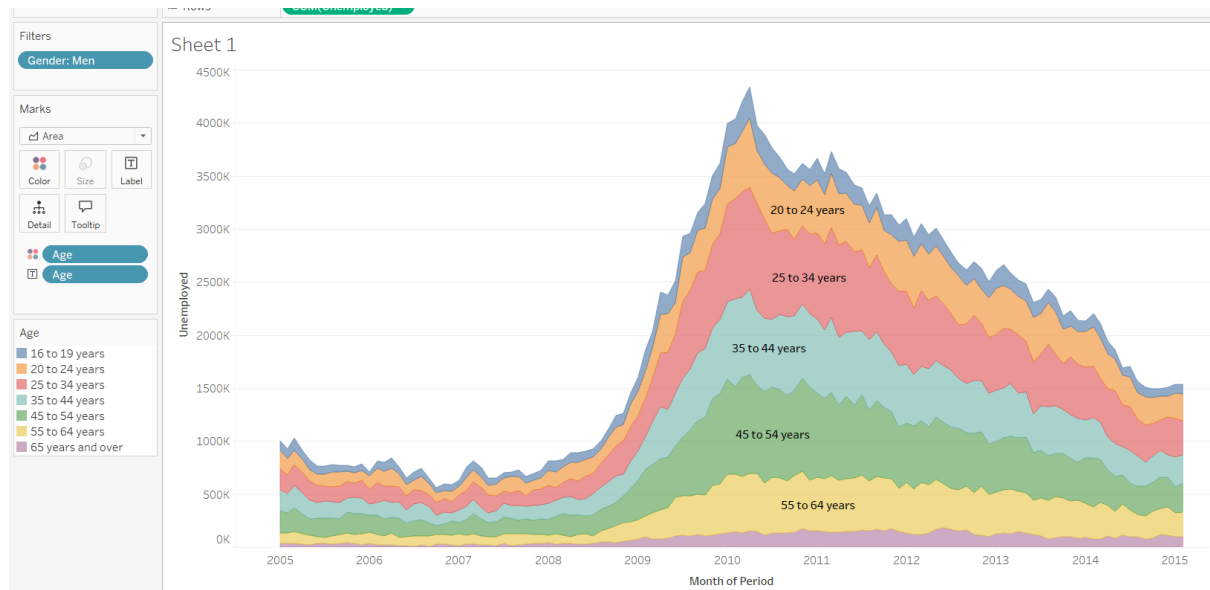


Format labels and make them bold.

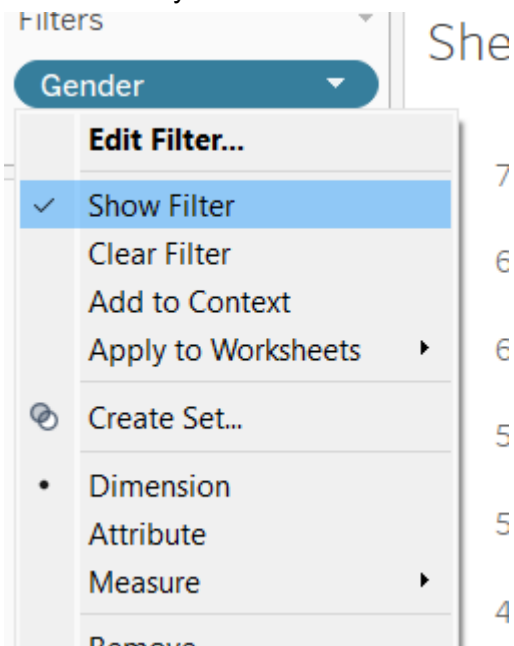
2) Adding a Filter and Quick Filter
Drag and drop Gender to the filter.



Now you can filter the data according to your condition.
Check the Men and Press OK.



To save time you can use the Show Filter option.



Change format of the label.

Open the file AmazingMartEU2.

Connections

Add

P1-AmazingMartEU2

Microsoft Excel

Sheets

ListOfOrders

OrderBreakdown

SalesTargets

New Union

New Table Extension

P1-AmazingMartEU2

Drag tables here to create a data model

[Learn more](#)

Drag ListOfOrders and OrderBreakdown here.
This will connect the data.

Connections

Add

P1-AmazingMartEU2

Microsoft Excel

Sheets

ListOfOrders

OrderBreakdown

SalesTargets

New Union

New Table Extension

ListOfOrders+ (P1-AmazingMartEU2)

Filters

0

Add

ListOfOrders

OrderBreakdown

ListOfOrd... — OrderBrea...

100 rows

How do relationships differ from joins? [Learn more](#)

ListOfOrders

Operator

OrderBreakdown

Order ID (OrderID)

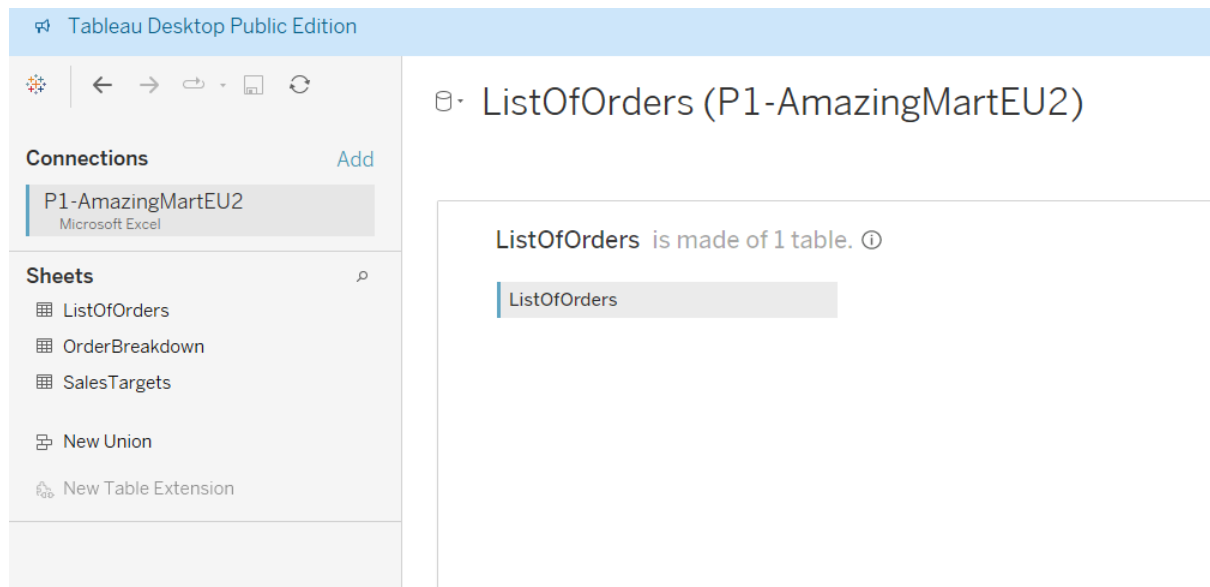
Add more fields

Performance Options

Remove Relationship

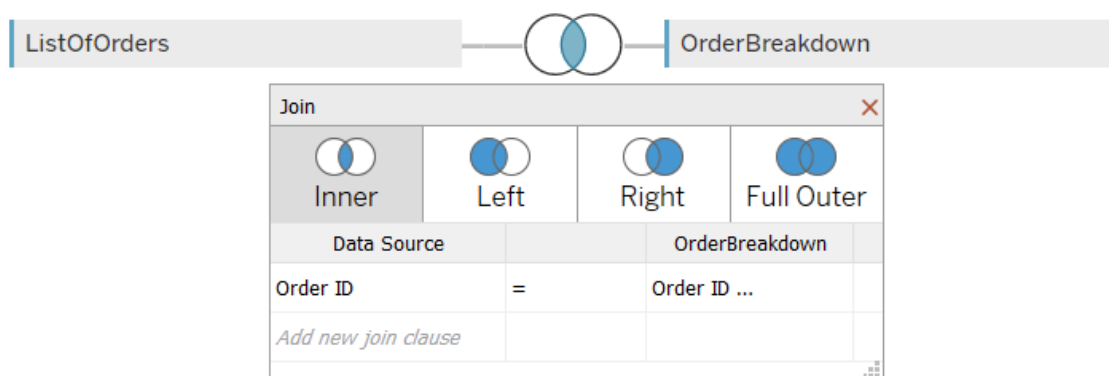
OrderBreakdown	OrderBreakdown	OrderBreakdown	OrderBreakdown	OrderBreakdown	OrderBreakdown
Order ID (OrderBreakdown)	Product Name	Discount	Sales	Profit	Quantity
BN-2011-7407039	Enermax Note Cards, Premium	0.500000	45	-26	3
AZ-2011-9050313	Dania Corner Shelving, Tradit...	0.000000	854	290	7
AZ-2011-6674300	Binney & Smith Sketch Pad, ...	0.000000	140	21	3
BN-2011-2819714	Boston Markers, Easy-Erase	0.500000	27	-22	2
BN-2011-2819714	Eldon Folders, Single Width	0.500000	17	-1	2
AZ-2011-617423	Binney & Smith Pencil Sharp...	0.000000	90	21	3

Remove OrderBreakdown.
Double click ListOfOrders.



Drag and Drop OrderBreakdown.

ListOfOrders is made of 2 tables. ⓘ



b) Joining Data in Tableau

Look at the data in excel.

Three Sheets are there.

We need to join these two data ListOfOrders and OrderBreakdown.

Drag ListOfOrders.

Same for OrderBreakdown.

Now again join these two.

Change name .

ListOfOrders+ OrderBreakdown

ListOfOrders is made of 2 tables. ⓘ



Click on Sheet1.

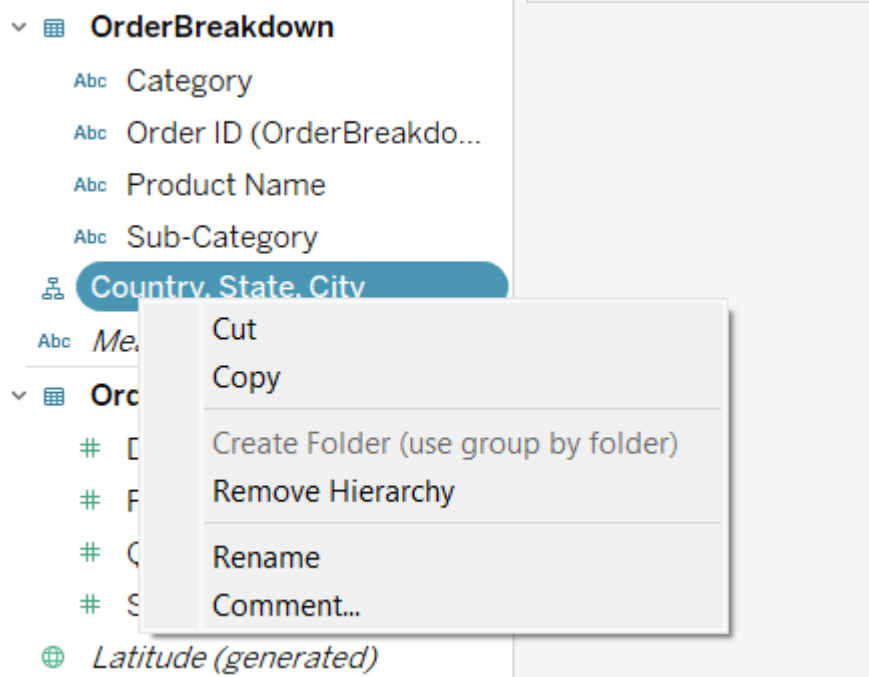
There are different types of data.

Some fields are picked as geographical data represented by the globe, some as text and some as date.

There are three elements of geographical data recognized by Tableau.

There is also hierarchy.

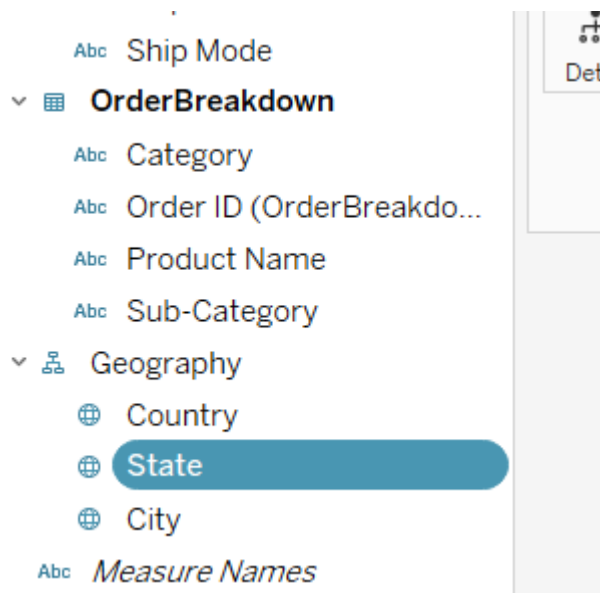
Remove hierarchy prompted by Tableau.



Drag City of ListOfOrders to Country.

Rename the Hierarchy as Geography.

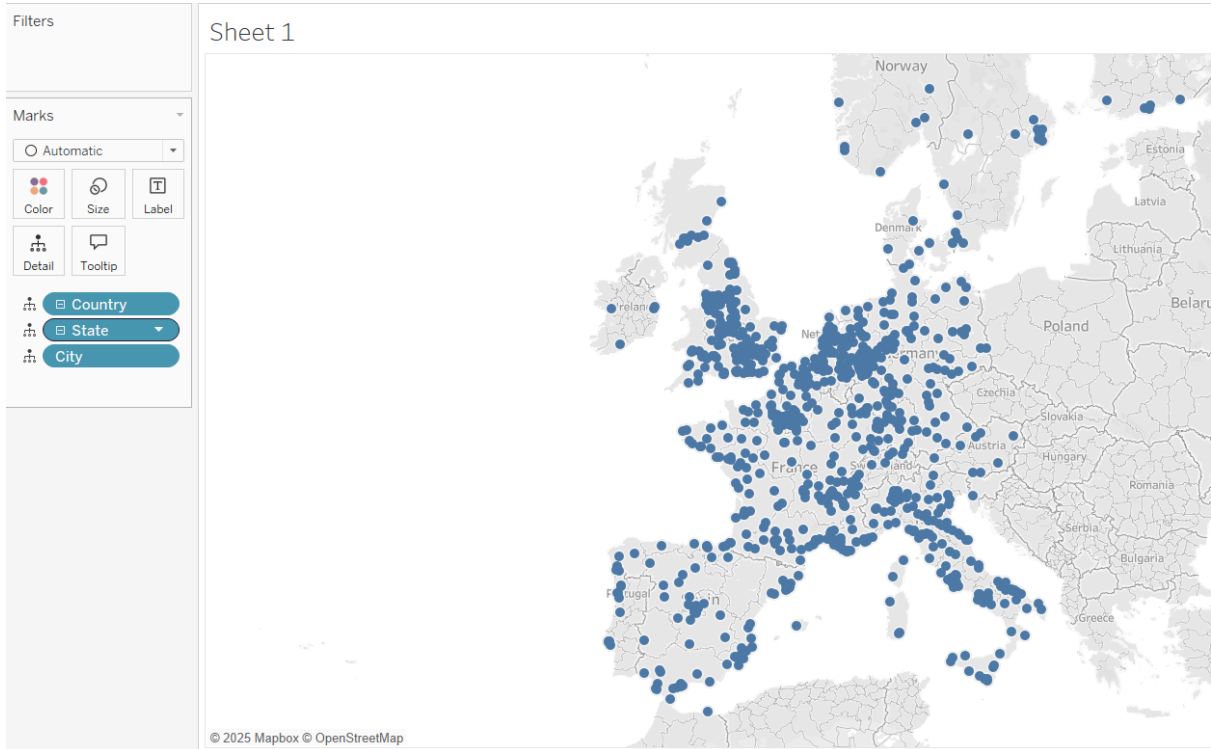
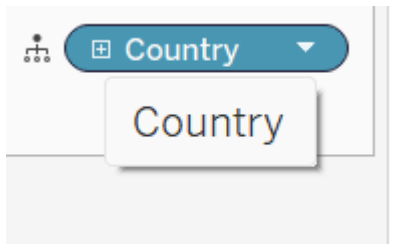
Move State in between Country and City.



Drag and drop Country to workspace.
Map of europe is seen.



Click this + sign to drill in further.

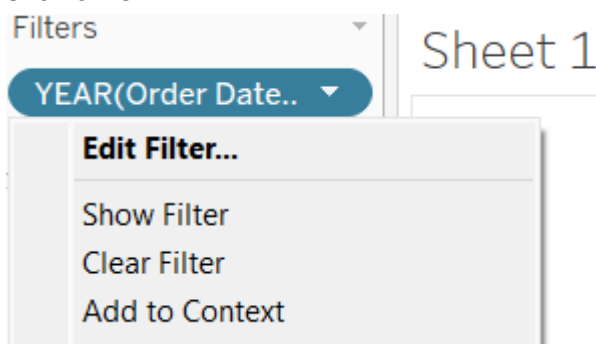


Take Order Date and Drag it into Filters.

Choose Years.

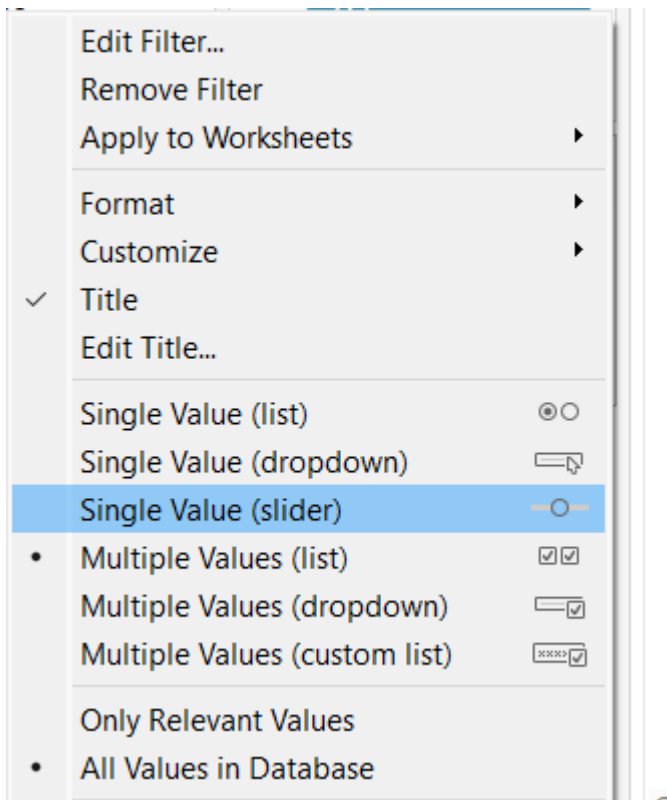
Click on 2012.

Click on Ok.

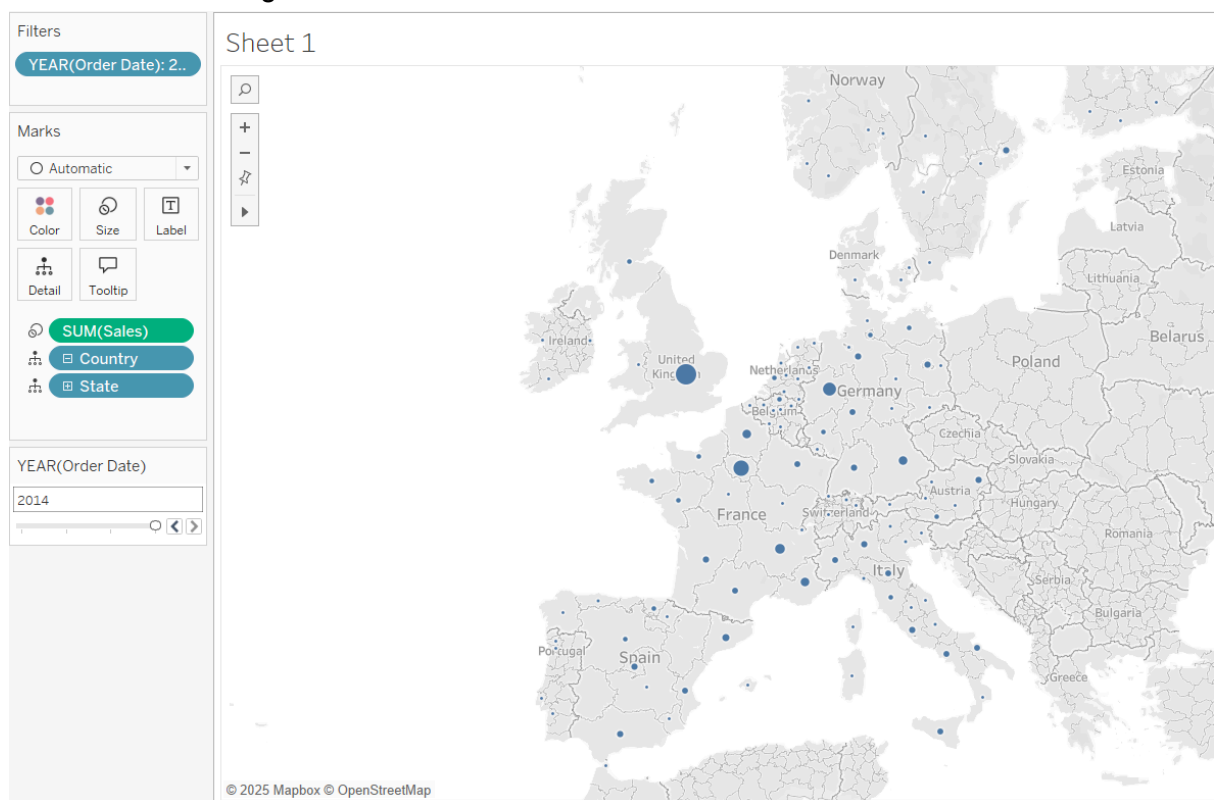


Click on Show Filter.

Change it to Single Value Slider.



Take Sales and Drag it to Size.



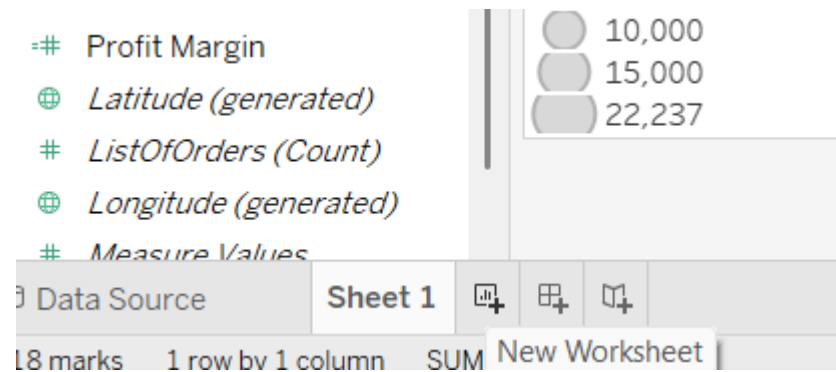
Give Black Border to the Sizes.

Right Click on Profit → Create → Calculated Field.

Take Profit Margin and Drag it into Color.

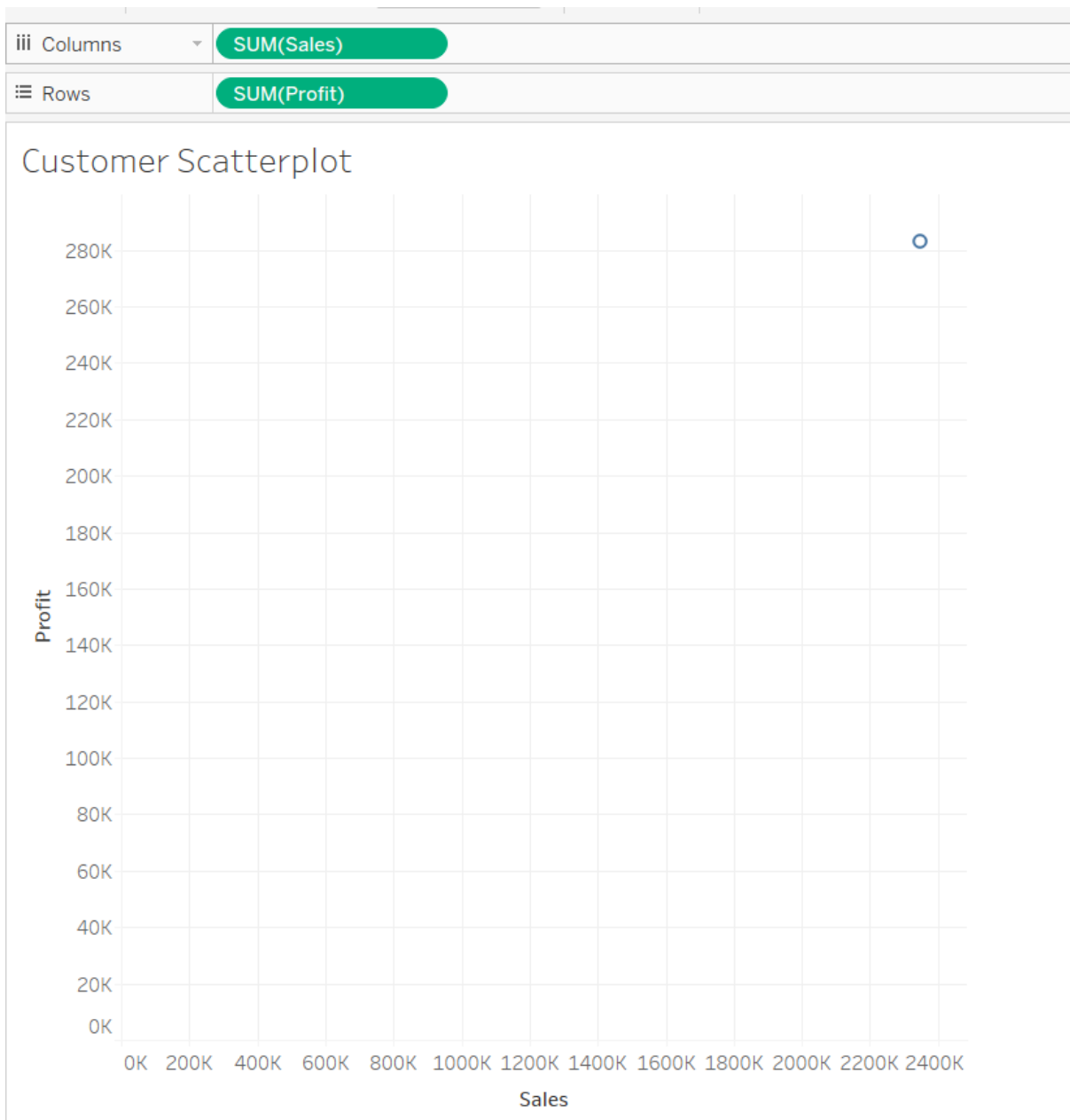
C) Creating a Scatter Plot, Applying Filters to Multiple Worksheets

Click on this button to create new worksheet.

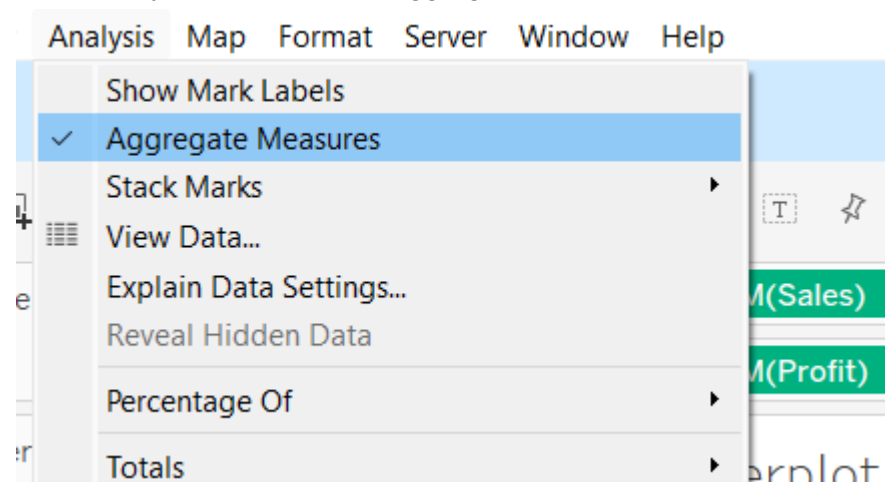


Rename Sheets as Map of Europe and Customer Scatter Plot.

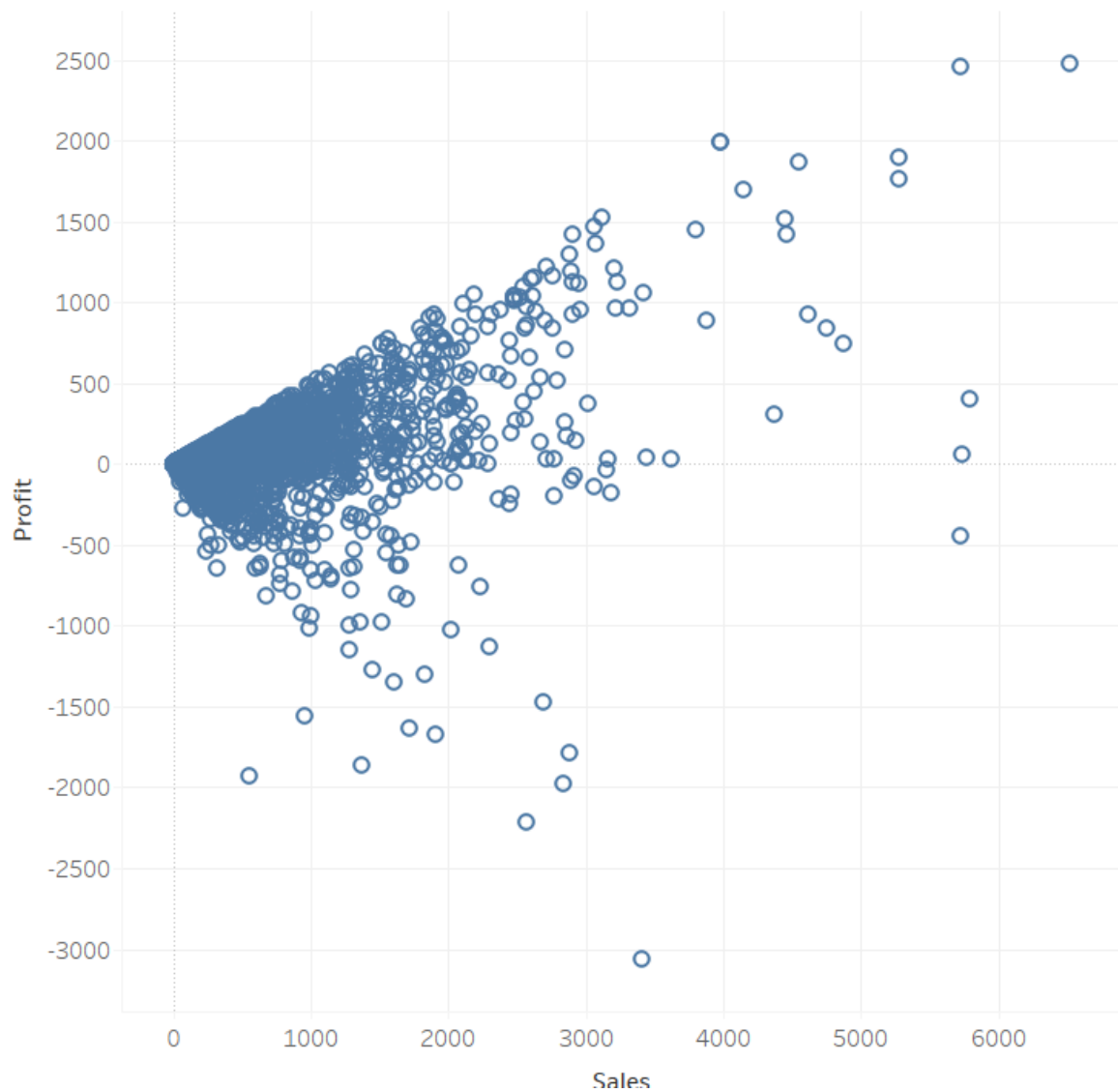
Drag and Drop Sales to Columns and Profit to Rows.



Go to Analysis and Uncheck Aggregate Measures.



Customer Scatterplot



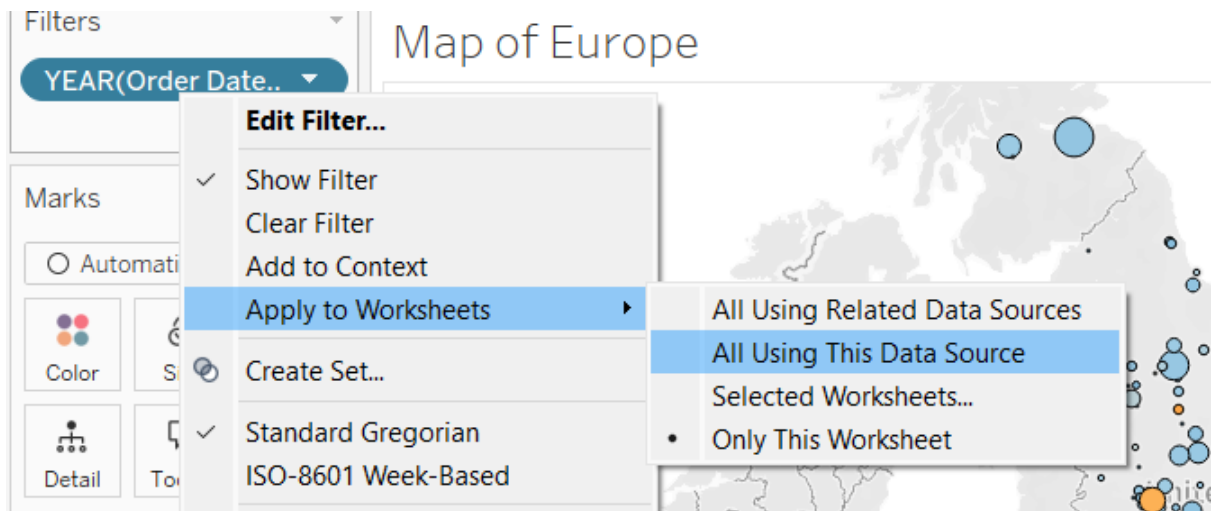
Again turn it on.

Drag and Drop Customer Name in Detail.

Work to create filters across multiple worksheets.

Go to the Map of Europe.

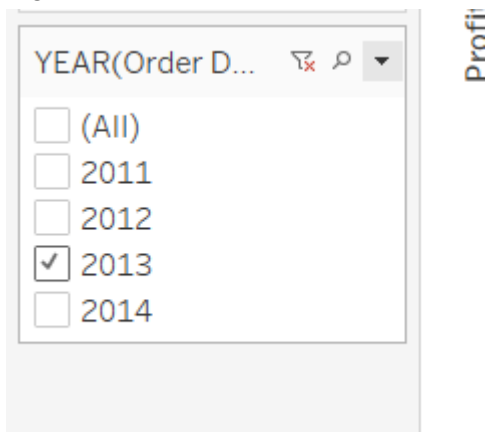
Right click on Filters.



This means any worksheet using this data source will automatically have this filter applied to it.

If you go to scatterplot you will see this filter has been applied already.

Right click on the filter of Customer Scatterplot and Click on Show Filter.



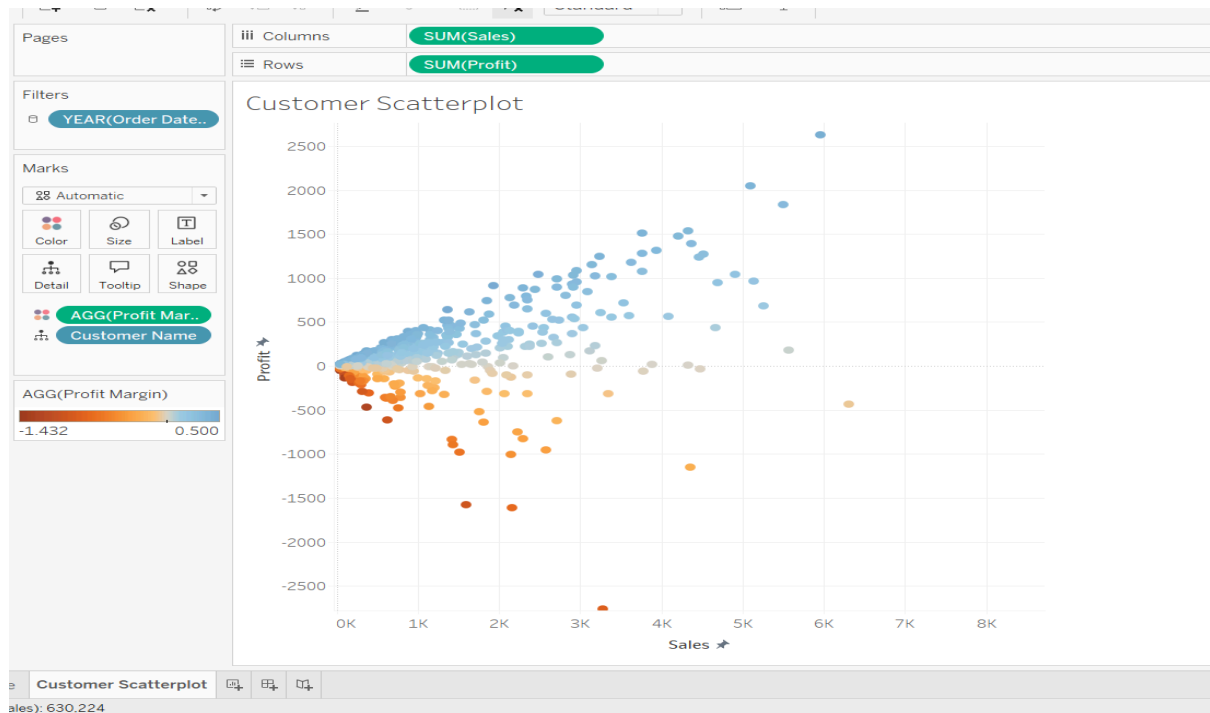
Customize → Show Values → Switch this off.

Change to Single Value Slider.

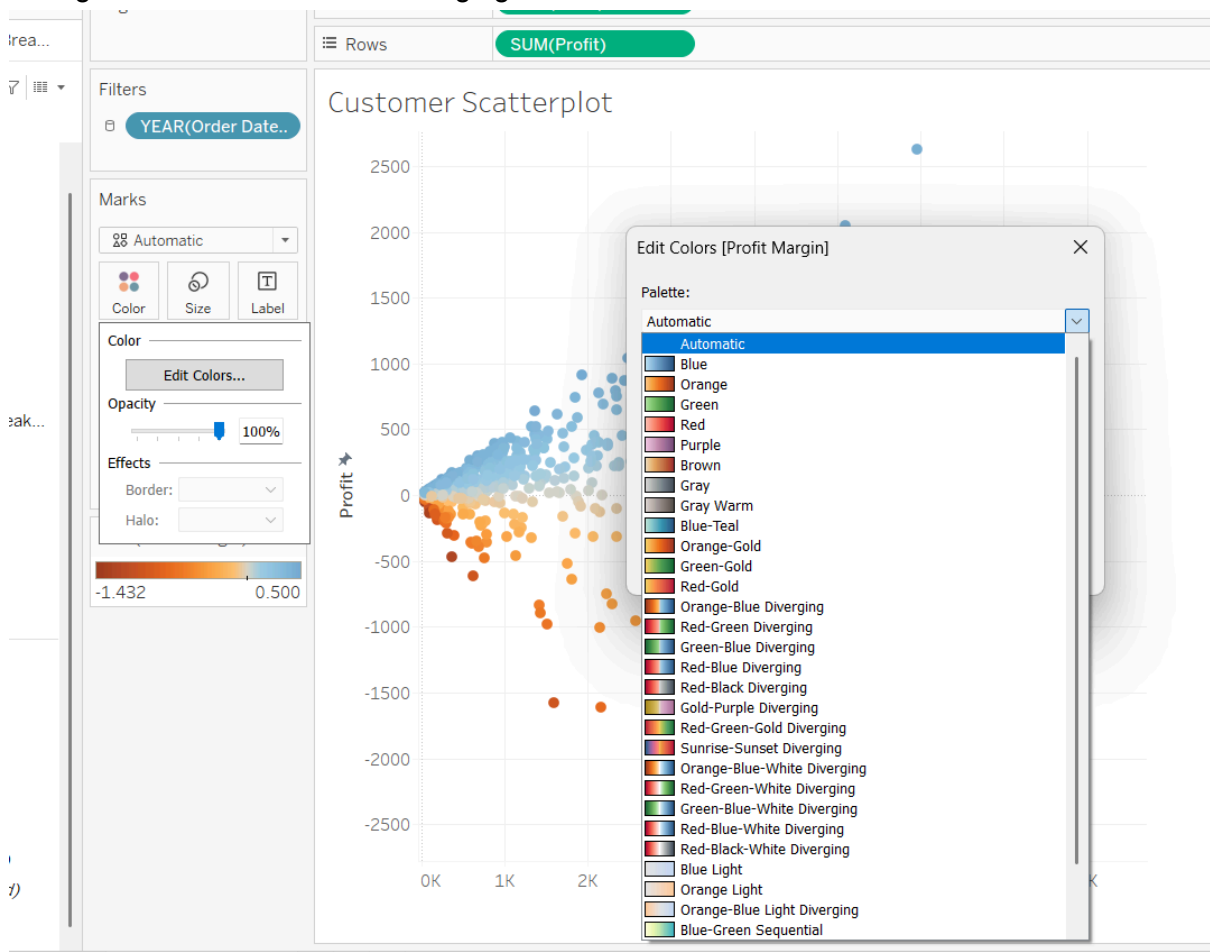
Drag and Drop ListOfOrders in New Sheet.

Delete it.

Drag and Drop ProfitMargin to Color on Scatter Plot.



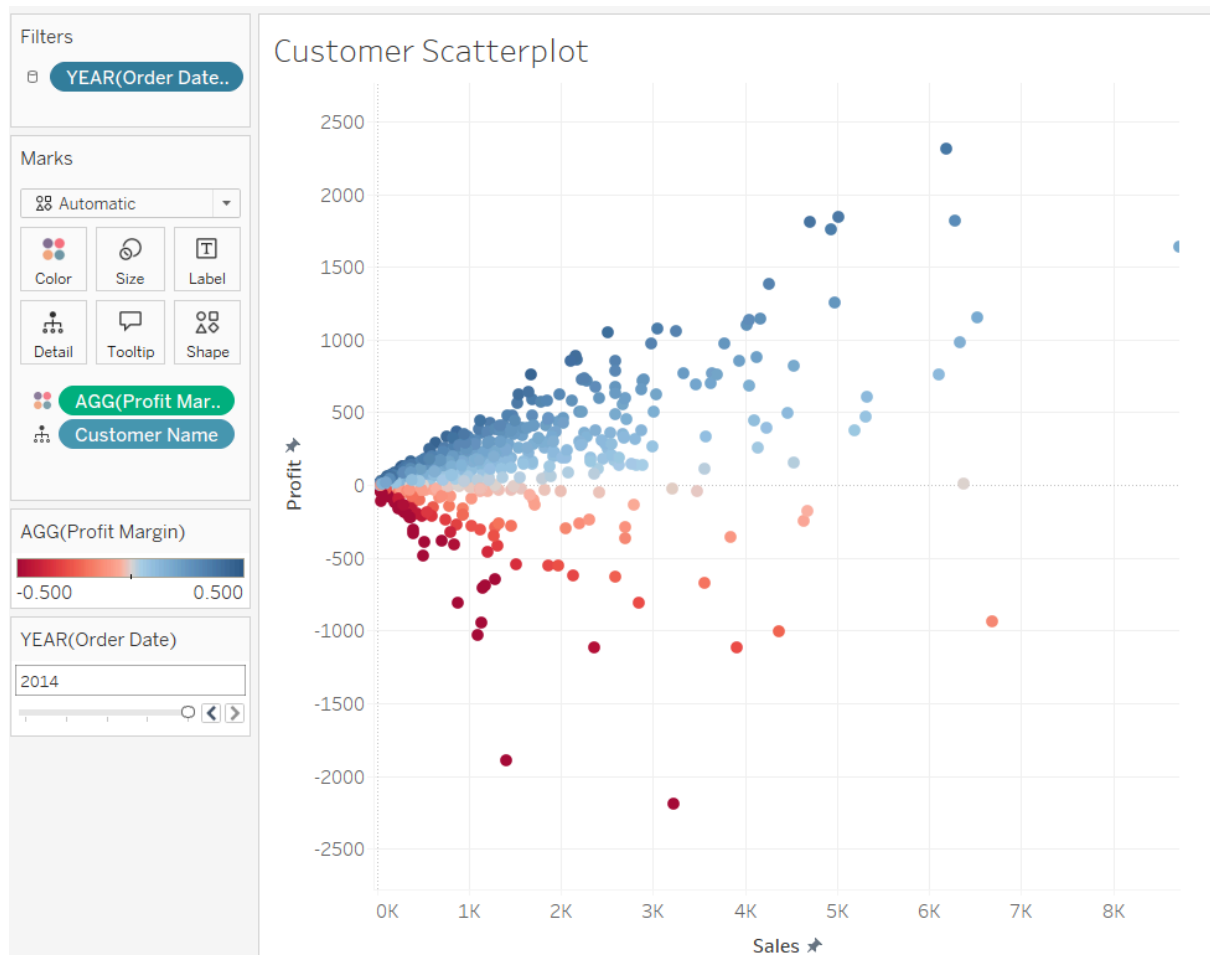
Change the shape as o.
Change color to red and blue diverging.



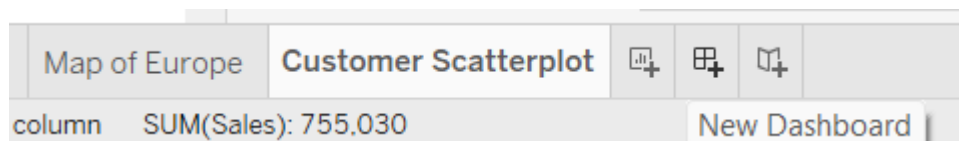
Change the size of the Bubble.

Edit Colors Start -0.5 and End 0.5.
Color Opacity 75%.

6) Creating Dashboard.

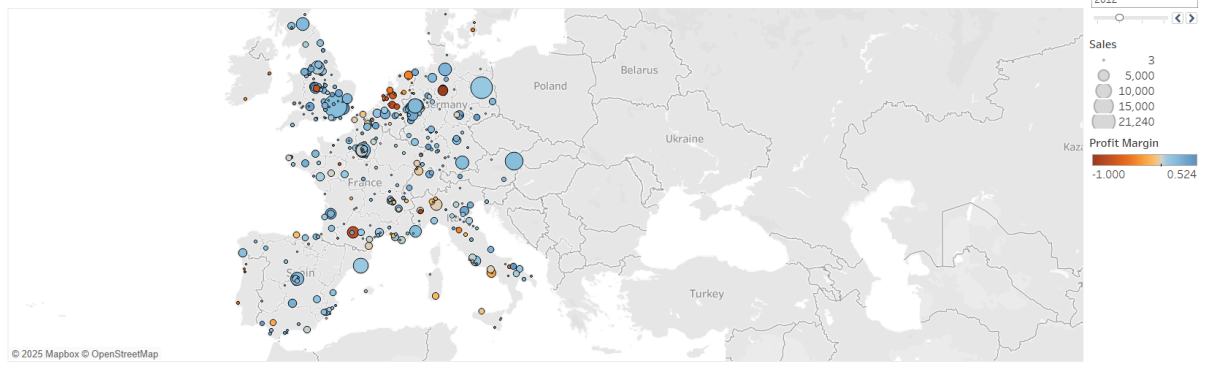


Click on New Dashboard.

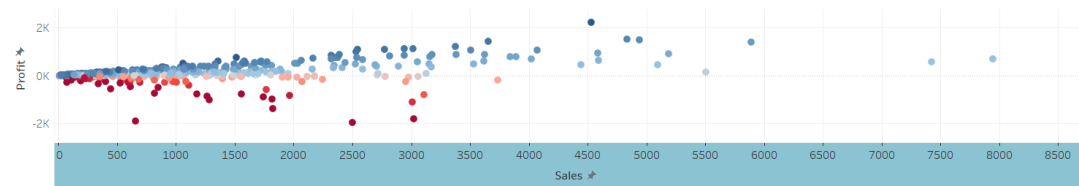


Drag and Drop Worksheets on Dashboard.
Drag and Drop both Map of Europe and Customer Scatter Plot.

Map of Europe



Customer Scatterplot



Apply filter.

SHOW ME

Year of Order Date

2012

Go to Sheet

Duplicate Sheet

✓ Title

Caption

Accessibility

Legends ▶

Filters ▶

Highlighters ▶

Show Page Control

View Toolbar ▶

✓ Use as Filter

Ignore Actions

Add Show/Hide Button

Floating

Select Container: Tiled

Deselect

Remove from Dashboard

Rename Dashboard Item...

Disable Auto Update

City

Country

Latitude (generated)

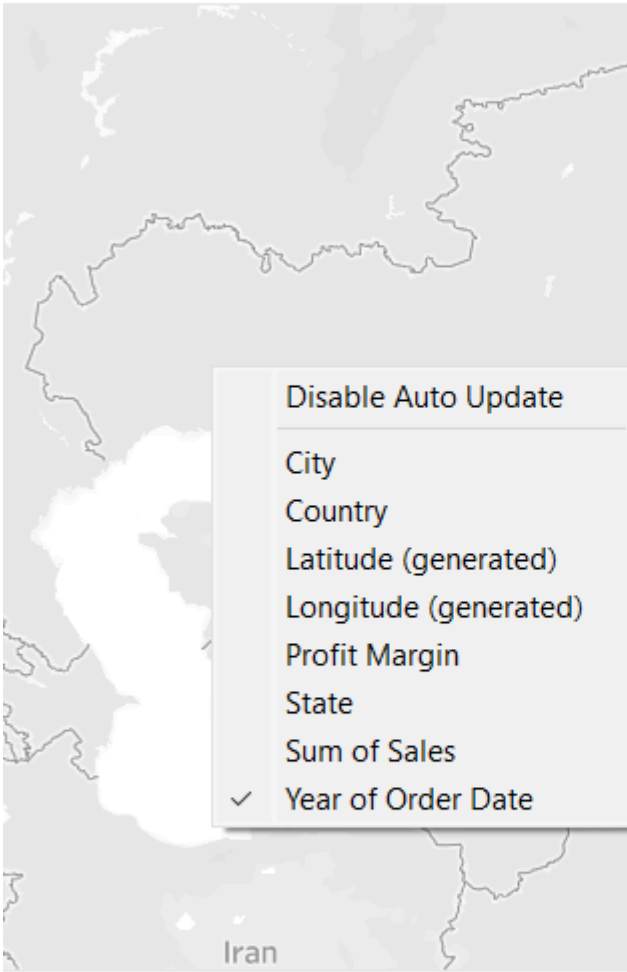
Longitude (generated)

Profit Margin

State

Sum of Sales

✓ Year of Order Date



Iran

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